

Technical Specification and Systems Engineering Blueprint for the Static Holographic Boundary Theory Synthetic Gravitational Lensing Telescope (SGLT)

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Abstract—The Static Holographic Boundary Theory Synthetic Gravitational Lensing Telescope (SGLT) is specified as a distributed deep-space observatory that replaces a conventional primary mirror with an electronically synthesized metric perturbation. The lensing model couples weak-field general relativity to boundary conformal-field-theory state congestion, assigning an effective ghost-seed mass to controlled register overflow and producing a tunable Schwarzschild deflection field. At the reference operating point, a seed mass of 10^{-6} solar masses establishes a 169.30-m focal baseline for a 1.000-m impact parameter. Metric projection is implemented by a cryogenic 8×8 InP/InGaAs heterojunction-bipolar-transistor array with niobium routing, lithographic airbridges, sapphire acoustic transport, and a nanoporous-silica impedance transformer coupled to liquid helium-4. Continuous entropic-debt compensation requires 906.00 kW; 1,786 lattice-assisted nuclear reaction (LANR) modules deliver 906.073 kW net while rejecting 4.6188 MW through 683.17 square metres of radiating surface at 600 K. A Lens Craft generates and stabilizes the artificial seed, while a Sensor/Focal Craft maintains the optical caustic and performs spectrographic acquisition. Dual-wavelength heterodyne metrology provides $0.17\text{-pm}/\sqrt{\text{Hz}}$ single-axis range noise and 15-nrad attitude knowledge. At Sun–Earth L2, differential solar-radiation pressure is rejected by a two-tier controller combining EMI–BF₄ electrospray station keeping with electro-optic phase correction through 100 kHz. Loss of LANR modules automatically derates the seed from 10^{-6} to 10^{-7} solar masses and repositions the Sensor/Focal Craft from 169.30 m to 1.693 km. Deterministic execution is provided by the bare-metal `shbt-os` runtime, a statically allocated 2,112-byte state frame, memory-mapped hardware control, and a six-cycle AVX-512 protection loop. The architecture enforces eigenvector detuning below 10^{-12} , commands an emitter-current shunt within 2.50 ns, and restores phase-locked operation within 120 ns. This specification closes the governing optical, electrical, thermal, structural, computational, navigation, and fault-containment interfaces required for an integrated preliminary design baseline.

Index Terms—artificial gravitational lensing, boundary conformal field theory, formation flight, heterojunction bipolar transistor, InP/InGaAs, lattice-assisted nuclear reaction, metric engineering, Sun–Earth L2, synthetic aperture.

I. MISSION ARCHITECTURE AND SYSTEM OVERVIEW

A. Mission-Level Physical Baseline

The SGLT shall synthesize an effective point-mass perturbation by commanding the boundary-state excess ΔN above

the admissible local-state limit. The reference coupling law, seed operating point, and associated uncertainty allocation are

$$\begin{aligned} M_{\text{seed}} &= \alpha_{\text{seed}} \Delta N, \\ \Delta N &= N_{\text{local}} - N_{\text{limit}}, \\ \alpha_{\text{seed}} &= 1.3258(13) \times 10^{-51} M_{\odot} \text{ bit}^{-1}, \\ M_{\text{seed},0} &= 1.0000(10) \times 10^{-6} M_{\odot}, \\ \Delta N_0 &= 7.542616(76) \times 10^{44} \text{ bit}. \end{aligned} \quad (1)$$

With the CODATA gravitational constant, defined speed of light, and nominal solar mass,

$$\begin{aligned} G &= 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ c &= 299\,792\,458 \text{ m s}^{-1}, \\ M_{\odot} &= 1.98847(7) \times 10^{30} \text{ kg}, \end{aligned} \quad (2)$$

the effective Schwarzschild radius and thin-lens focal distance are

$$\begin{aligned} R_{s,0} &= \frac{2GM_{\text{seed},0}}{c^2} = 2.95334(11) \times 10^{-3} \text{ m}, \\ f_0 &= \frac{r_0^2}{2R_{s,0}}, \\ r_0 &= 1.0000(5) \text{ m}, \\ f_0 &= 169.30(19) \text{ m}. \end{aligned} \quad (3)$$

The mission-level stability condition is imposed directly on the holographic transport eigenvector:

$$|\delta\mu| \leq \delta\mu_{\text{floor}} = 1.0 \times 10^{-12}. \quad (4)$$

The boundary-condition data set declares a bit-congestion radius and an operational exclusion radius as

$$\begin{aligned} R_{\text{congestion,decl}} &= 2.954 \times 10^{15} \text{ m} = 1.9746 \times 10^4 \text{ AU}, \\ R_{\text{excl,op}} &= 19.7 \text{ AU} = 2.9471 \times 10^{12} \text{ m}, \\ 1 \text{ AU} &= 1.495978707 \times 10^{11} \text{ m}. \end{aligned} \quad (5)$$

Equation (5) prevents the dimensionally inconsistent identification of $2.954 \times 10^{15} \text{ m}$ with 19.7 AU. The two values belong to distinct boundary and bulk domains and are related by the calibrated holographic scale factor developed in Section II-B; they are not alternate unit representations of one physical radius. Mission assurance shall apply the declared congestion radius to boundary-capacity and numerical-stability analyses and the 19.7-AU radius to commanded inter-seed separation. Sun–Earth L2 selection mitigates externally driven detuning

through low differential acceleration and continuous formation observability; it does not waive either radius constraint.

B. Two-Craft Formation and L2 Operations

The flight segment comprises a Lens Craft and a Sensor/Focal Craft. The Lens Craft carries the metric-field generator, 8×8 cryogenic transducer array, LANR electrical plant, helium-4 acoustic sink, radiator wings, deterministic control computer, and emergency current-diversion network. The Sensor/Focal Craft carries the photon-counting focal-plane assembly, spectrometers, wavefront sensor, optical crosslinks, and cold-gas/microelectric formation actuators. Their commanded relative state is

$$\begin{aligned} \mathbf{r}_{\text{SF/LC}} &= f_0 \hat{\mathbf{k}} + \delta \mathbf{r}, \\ f_0 &= 169.30 \text{ m}, \\ \|\delta \mathbf{r}\|_{3\sigma} &\leq 1.00 \times 10^{-9} \text{ m}, \\ \|\delta \mathbf{v}\|_{3\sigma} &\leq 5.00 \times 10^{-10} \text{ m s}^{-1}, \end{aligned} \quad (6)$$

where $\hat{\mathbf{k}}$ is the instantaneous optical-axis unit vector. The baseline Sun–Earth L2 halo orbit shall satisfy an annual station-keeping allocation below 2.5 m s^{-1} , compared with an atmospheric-drag-dominated requirement exceeding 150 m s^{-1} in low Earth orbit (LEO):

$$\begin{aligned} \Delta V_{\text{L2}} &< 2.5 \text{ m s}^{-1} \text{ yr}^{-1}, \\ \Delta V_{\text{LEO}} &> 150 \text{ m s}^{-1} \text{ yr}^{-1}. \end{aligned} \quad (7)$$

Table I defines the deployment trade using the formation dry mass of 5.370 Mg established in Table II, a ten-year service interval, a 30% maneuver reserve, and electric propulsion with a specific impulse of 1600 s. The site-trade reserve is computed independently of the installed 1 190-kg propellant load. The reserved propellant follows

$$\begin{aligned} m_{p,10} &= m_d \left[\exp \left(\frac{1.30 n \Delta V_{\text{yr}}}{g_0 I_{\text{sp}}} \right) - 1 \right], \\ m_d &= 5\,370 \text{ kg}, \quad n = 10 \text{ yr}, \\ g_0 &= 9.80665 \text{ m s}^{-2}, \quad I_{\text{sp}} = 1600 \text{ s}. \end{aligned} \quad (8)$$

C. Integrated Mass, Power, and Thermal Closure

The master resource allocation in Table II establishes the dry and wet masses, installed propellant, continuous bus power, maneuver capability, and telecommunications interfaces for both flight elements. All operational sites retain the same science hardware, power plant, and thermal rejection architecture.

The allocated maneuver capabilities are vehicle-level quantities and are not additive at formation level.

At the reference seed mass, entropic-debt power, reactor count, net generation, waste heat, and radiator area close as

$$\begin{aligned} P_{\text{debt}} &= \left(\frac{M_{\text{seed},0}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}) = 906.00 \text{ kW}, \\ N_{\text{LANR}} &= \left\lceil \frac{906\,000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil = 1786, \\ P_{\text{net}} &= 1786 (507.3195 \text{ W}) = 906.0726 \text{ kW}, \\ P_{\text{th}} &= 1786 (3093.44 \text{ W}) = 5.52488 \text{ MW}, \\ P_{\text{rej}} &= P_{\text{th}} - P_{\text{net}} = 4.61881 \text{ MW}. \end{aligned} \quad (9)$$

For emissivity $\epsilon = 0.92$, radiator temperature $T_r = 600 \text{ K}$, background temperature $T_b = 2.7 \text{ K}$, and Stefan–Boltzmann constant $\sigma = 5.670374419(86) \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, the two-sided effective radiator area is

$$\begin{aligned} A_r &= \frac{P_{\text{rej}}}{\epsilon \sigma (T_r^4 - T_b^4)} \\ &= 683.17 \text{ m}^2, \\ \delta A_r|_{\delta T_r = \pm 2 \text{ K}} &= \mp 9.11 \text{ m}^2. \end{aligned} \quad (10)$$

D. Transducer, Lithography, and Deterministic-Control Interfaces

The primary metric projector is a 64-channel InP/InGaAs HBT array on a $50.0 \mu\text{m}$ pitch. Table III is the controlling GDSII mask-interface definition; all coordinates shall be snapped to a 1.0-pm database grid.

The bare-metal `shbt-os` controller shall execute without dynamic allocation from a 2112-byte `UnifiedStinespringFrame`, partitioned into a 640-byte active workspace and a 1472-byte environmental-state ledger. The `SHBT-MMIO-1` interface occupies the physical base address `0x70000000` and implements the register map in Table IV.

At a 4.000-GHz safety clock, the six-cycle AVX-512 monitor has a nominal decision latency of 1.50 ns. A violation of Eq. (4) shall assert `SHUNT_TRIG`, reduce all HBT gate commands below the 3.800-V turn-on threshold, and initiate the recovery sequencer according to

$$\begin{aligned} t_{\text{SIMD}} &= \frac{6}{4.000 \times 10^9 \text{ s}^{-1}} = 1.50 \text{ ns}, \\ t_{\text{shunt}} &\leq 2.50 \text{ ns}, \\ t_{\text{recover}} &\leq 120 \text{ ns}, \\ V_{\text{gate},i}(t \geq t_{\text{shunt}}) &< 3.800 \text{ V}, \quad i \in \{0, 1, \dots, 63\}. \end{aligned} \quad (11)$$

The resulting baseline closes the mission geometry, formation-control, mass, electrical, thermal, lithographic, software, and hardware-protection interfaces required to enter preliminary design at Sun–Earth L2.

II. RELATIVISTIC GRAVITATIONAL LENSING AND MULTI-SEED WAVEFRONT OPTICS

A. Linearized ADM Metric and Mass–Congestion State Equation

The lensing field is represented in a $3+1$ Arnowitt–Deser–Misner (ADM) foliation with lapse α_{ADM} , shift β^i , and spatial metric γ_{ij} . The spacetime interval is

$$\begin{aligned} ds^2 &= -\alpha_{\text{ADM}}^2 c^2 dt^2 + \gamma_{ij} dX^i dX^j, \\ dX^i &= dx^i + \beta^i c dt, \\ \alpha_{\text{ADM}}^{(0)} &= 1, \quad \beta^{i(0)} = 0, \quad \gamma_{ij}^{(0)} = \delta_{ij}. \end{aligned} \quad (12)$$

TABLE I
DEPLOYMENT-SITE MASS, POWER, THERMAL, AND STABILITY TRADE

Deployment site	Disturbance environment	ΔV allocation (m/s/yr)	10-yr propellant (kg)	Wet mass (Mg)	Net / rejected power (MW)	Rigidity disposition
LEO, 550 km	Drag, eclipse cycling, Earth gravity gradient, atomic oxygen	> 150	> 711	> 6.081	0.9109/4.6188	Noncompliant; continuous drag and gravity-gradient forcing exceed the eigenvector disturbance allocation.
Earth–Moon L5	Earth–Moon gradient, solar pressure, weak long-period libration	35	158	5.528	0.9109/4.6188	Marginal; periodic differential acceleration consumes the formation and rigidity margins.
Sun–Earth L2 halo	Solar pressure, weak halo-orbit instability, no atmospheric drag	< 2.5	< 11.2	< 5.382	0.9109/4.6188	Baseline; full compliance with Eq. (4) under closed-loop metrology.
Heliocentric, 1.1 AU	Solar pressure and gravity; increased communications latency	< 0.5	< 2.3	< 5.373	0.9109/4.6188	Compliant; lowest maneuver load, with reduced command and ranging availability.

TABLE II
MASTER SPACECRAFT RESOURCE ALLOCATION

Element	Dry mass (kg)	Propellant (kg)	Wet mass (kg)	Propellant system	Bus power (kW)	Δv (m/s)	Telecommunications
Lens Craft	4 250.0	850.0	5 100.0	EMI–BF ₄ electro-spray / N ₂	906.0726	1 450.0	25.0 Mbps S/Ka-band
Sensor Craft	1 120.0	340.0	1 460.0	N ₂ and microelectric formation propulsion	4.80	2 100.0	12.0 Gbps laser optical downlink
Total system	5 370.0	1 190.0	6 560.0	Dual-craft allocation	910.8726	Vehicle-specific	S/Ka command plus optical science downlink

TABLE III
CONTROLLING GDSII LAYER DEFINITION

Layer	Purpose code	Material	Thickness / geometry
10	SUBSTRATE_INP	SI InP (100)	350.0 μm wafer
20	AIRBRIDGE_SPAN	Au/InP	1.5 μm $\times$ 5.0 μm span
25	MET_NB_TRACE	Nb	300.0 nm; $T_c = 9.20$ K

For a static, weak perturbation in isotropic coordinates and metric signature $(-, +, +, +)$, the synthesized Newtonian potential and first-order metric are

$$\begin{aligned}
 \Phi(r) &= -\frac{GM_{\text{seed}}}{r}, \quad r = \sqrt{x^2 + y^2 + z^2}, \\
 g_{\mu\nu} &= \eta_{\mu\nu} + h_{\mu\nu} + \mathcal{O}(h^2), \\
 \eta_{\mu\nu} &= \text{diag}(-1, 1, 1, 1), \\
 h_{00} &= -\frac{2\Phi}{c^2} = \frac{R_s}{r}, \\
 h_{0i} &= 0, \\
 h_{ij} &= -\frac{2\Phi}{c^2} \delta_{ij} = \frac{R_s}{r} \delta_{ij}.
 \end{aligned} \tag{13}$$

The mass–congestion constitutive equation maps the localized state excess directly to the effective lens mass:

$$\begin{aligned}
 M_{\text{seed}} &= \alpha_{\text{seed}} (N_{\text{local}} - N_{\text{limit}}) = \alpha_{\text{seed}} \Delta N, \\
 \alpha_{\text{seed}} &= 1.3258 \times 10^{-51} M_{\odot} \text{ bit}^{-1}, \\
 \Delta N &= \frac{M_{\text{seed}}}{\alpha_{\text{seed}}}.
 \end{aligned} \tag{14}$$

The associated gravitational length scale is obtained by equating the weak-field coefficient to the Schwarzschild perturbation:

$$\begin{aligned}
 R_s &= \frac{2GM_{\text{seed}}}{c^2} = \frac{2G\alpha_{\text{seed}}\Delta N}{c^2}, \\
 \hat{\alpha}_{\text{GR}}(r_0) &= \frac{4GM_{\text{seed}}}{c^2 r_0} = \frac{2R_s}{r_0}.
 \end{aligned} \tag{15}$$

Under the paraxial approximation, a ray incident parallel to the optical axis intersects that axis at

$$\begin{aligned}
 f(r_0) &= \frac{r_0}{\hat{\alpha}_{\text{GR}}(r_0)} = \frac{r_0^2}{2R_s} = \frac{c^2 r_0^2}{4GM_{\text{seed}}}, \\
 r_0 &= 1.0000 \text{ m}.
 \end{aligned} \tag{16}$$

(13) The complete commanded operating range is given in Table V. Entropy-debt power follows the prescribed linear mass law.

$$P_{\text{debt}} = \left(\frac{M_{\text{seed}}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}). \tag{17}$$

TABLE IV
SHBT-MMIO-1 FLIGHT-CONTROL REGISTER MAP

Address	Symbol	Access	Width	Required function
0x70000000	CR0_CTRL	R/W	32 bit	Enable, operating-mode select, phase-loop arm, and synchronous soft reset.
0x70000004	SR0_STAT	R	32 bit	Quench-detect, phase-lock, thermal-alarm, metrology-valid, and shunt-latched status.
0x70000008	DET_MU_HI	R	32 bit	Upper word of the signed Q1.62 eigenvector-detuning telemetry value.
0x7000000C	DET_MU_LO	R	32 bit	Lower word of the signed Q1.62 eigenvector-detuning telemetry value.
0x70000010	BIT_OVF_N	R/W	32 bit	Upper command word of the 64-bit state-overflow mantissa.
0x70000014	BIT_OVF_L	R/W	32 bit	Lower command word of the 64-bit state-overflow mantissa.
0x70000018	RF_PHASE_V	R/W	32 bit	Sixteen-bit channel address and sixteen-bit DAC code spanning 3.800–7.400 V.
0x7000001C	SHUNT_TRIG	W	32 bit	Nonzero write directly fires the solid-state emitter-current diversion network.
0x70000020	REC_SEQ_PC	R/W	32 bit	Recovery entry-point command, microsequence program counter, and terminal-state code.
0x70000040	INT_TENS_0	R/W	512 bit	Base of the 64-byte interference-tensor command aperture.

TABLE V
SYNTHETIC-SEED LENSING AND ENTROPY-DEBT OPERATING REGIMES FOR $r_0 = 1.0000$ m

$M_{\text{seed}}/M_{\odot}$	State overflow ΔN (bit)	R_s (m)	Focal distance f	P_{debt}
1.0×10^{-7}	7.542616×10^{43}	2.953339×10^{-4}	1.692999 km	90.60 kW
1.0×10^{-6}	7.542616×10^{44}	2.953339×10^{-3}	169.2999 m	906.00 kW
1.0×10^{-5}	7.542616×10^{45}	2.953339×10^{-2}	16.92999 m	9.060 MW
1.0×10^{-4}	7.542616×10^{46}	2.953339×10^{-1}	1.692999 m	90.60 MW
1.0×10^{-3}	7.542616×10^{47}	2.953339	0.1692999 m	906.0 MW

The tabulated bit counts are rounded to seven significant digits. Their relative quantization and constant-propagation bounds are

$$\begin{aligned} \left| \frac{\delta(\Delta N)}{\Delta N} \right| &\leq 5.0 \times 10^{-7}, \\ \left| \frac{\delta R_s}{R_s} \right| &\leq \sqrt{\left(\frac{\delta G}{G} \right)^2 + \left(\frac{\delta M_{\text{seed}}}{M_{\text{seed}}} \right)^2}, \\ \left| \frac{\delta f}{f} \right| &\leq 2 \left| \frac{\delta r_0}{r_0} \right| + \left| \frac{\delta R_s}{R_s} \right|. \end{aligned} \quad (18)$$

B. Theoretical Closure of Bit-Congestion vs. Exclusion Radius

The declared asymptotic bit-congestion radius and the bulk operational exclusion radius are coordinates in different domains. Their common-unit values and direct dimensionless map are

$$\begin{aligned} R_{\text{congestion,decl}} &= 2.954 \times 10^{15} \text{ m} = 0.09573 \text{ pc}, \\ R_{\text{excl,op}} &= 19.7 \text{ AU} = 2.9471 \times 10^{12} \text{ m}, \\ \chi_{\text{SHBT}} &\equiv \frac{R_{\text{congestion,decl}}}{R_{\text{excl,op}}} = 1.002341 \times 10^3, \\ R_{\text{congestion,decl}} &= \chi_{\text{SHBT}} R_{\text{excl,op}}. \end{aligned} \quad (19)$$

Thus, χ_{SHBT} is a boundary-to-bulk coordinate calibration, not an astronomical-unit conversion. In particular, 2.954×10^{15} m is 0.09573 pc; a value of 0.312 pc would instead be approximately 9.63×10^{15} m.

The proposed quantum-gravity combination is dimensionless, but its numerical value at the reference seed mass does

not by itself produce the scale in Eq. (19). Direct substitution gives

$$\begin{aligned} \chi_{\text{QG}} &\equiv \left(\frac{\hbar c}{GM_{\text{seed},0}^2} \right)^{-1/4}, \\ M_{\text{seed},0} &= 10^{-6} M_{\odot}, \\ \chi_{\text{QG}} &= 9.5584 \times 10^{15}. \end{aligned} \quad (20)$$

Consequently, a dimensionless SHBT matching coefficient is required for a mathematically closed constitutive map:

$$\begin{aligned} \chi_{\text{SHBT}} &= C_{\text{SHBT}} \left(\frac{\hbar c}{GM_{\text{seed},0}^2} \right)^{-1/4}, \\ C_{\text{SHBT}} &\equiv \frac{\chi_{\text{SHBT}}}{\chi_{\text{QG}}} = 1.04865 \times 10^{-13}. \end{aligned} \quad (21)$$

Equation (21) separates the postulated mass-dependent quantum factor from the empirically declared boundary-to-bulk calibration. Setting $C_{\text{SHBT}} = 1$ would contradict the two stated radii by more than twelve orders of magnitude.

For K simultaneously active seeds, the metric is decomposed into isolated weak-field responses and the non-additive overlap correction:

$$\begin{aligned} g_{\mu\nu}(x) &= \eta_{\mu\nu} + \sum_{i=1}^K h_{\mu\nu}^{(i)}(x) + \delta g_{\mu\nu}^{\text{overlap}}(x), \\ \delta g_{\mu\nu}^{\text{overlap}}(x) &= \sum_{i < j} \mathcal{C}_{\mu\nu}^{\alpha\beta\gamma\delta} h_{\alpha\beta}^{(i)}(x) h_{\gamma\delta}^{(j)}(x) W\left(\frac{d_{ij}}{R_{\text{excl,op}}}\right), \\ W(q) &= 0 \quad \text{for } q \geq 1, \end{aligned} \quad (22)$$

where $\mathcal{C}$ is the configuration-controlled SHBT coupling tensor, d_{ij} is inter-seed separation, and W is the qualified overlap

window. Let K_0 denote the nominal holographic transport operator, $\delta K[\delta g^{\text{overlap}}]$ its overlap-induced perturbation, and V the eigenvector matrix of K_0 . First-order eigenvalue perturbation is bounded by

$$\begin{aligned} |\delta \mu_a| &\leq \kappa_2(V) \|\delta K[\delta g^{\text{overlap}}]\|_2, \\ \max_a |\delta \mu_a| &\leq \delta \mu_{\text{floor}} = 1.0 \times 10^{-12}. \end{aligned} \quad (23)$$

The flight computer shall therefore inhibit a second seed unless both $d_{ij} \geq R_{\text{excl,op}}$ and the bound in Eq. (23) is satisfied. An exclusion-radius violation, invalid scale-map coefficient, inconsistent seed ephemeris, or $|\delta \mu| > 10^{-12}$ asserts the hardware shunt predicate in Eq. (88); the deterministic sequence in Eq. (89) then either restores an ECC-clean, geometry-valid state or leaves the metric field inhibited.

C. Multi-Seed Superposition and the 512-Bit Interference Tensor

A single monopolar seed has an inverse focal power proportional to r_0^{-2} and therefore produces a longitudinal caustic rather than a planar focal surface. The controlled multi-seed metric is

$$g_{\mu\nu} = \eta_{\mu\nu} + \sum_{i=1}^K h_{\mu\nu}^{(i)} + I_{\mu\nu}, \quad (24)$$

where $I_{\mu\nu}$ is the actively commanded interference tensor. The uncorrected radial focal power is

$$\begin{aligned} \mathcal{P}_r(r_0) &= \frac{1}{f(r_0)} = \frac{2R_s}{r_0^2}, \\ \frac{d\mathcal{P}_r}{dr_0} &= -\frac{4R_s}{r_0^3}. \end{aligned} \quad (25)$$

For reference aperture radius $r_{\text{ref}} = 1.0000$ m, the diagonal compensation command is defined by

$$\begin{aligned} I_{00}(r_0) &= -\frac{R_s}{r_0} + \frac{2R_s r_0}{r_{\text{ref}}^2}, \\ I_{11}(r_0) &= I_{22}(r_0) = I_{33}(r_0) = I_{00}(r_0), \\ I_{\mu\nu}(r_0) &= 0, \quad \mu \neq \nu. \end{aligned} \quad (26)$$

At the $10^{-6} M_\odot$ operating point and $r_0 = r_{\text{ref}}$, all four diagonal coefficients reduce to one 512-bit binary value. The rug/MPFR representation, evaluated with round-to-nearest and ties-to-even, is specified exactly by the sign-mantissa-exponent tuple

$$\begin{aligned} \mathcal{I}_{512} &= (+1) 0x\text{C18CD00F59F01543276459773BBF9420} \\ &\quad 079F061062F313216FF7064EAF5C3D2 \\ &\quad 9C7248835CEF0BB3C726C4C9A51009A3 \\ &\quad 089BDAF38750E9B1523880972C33DD05 \times 2^{-520}. \end{aligned} \quad (27)$$

Its exact 155-significant-digit decimal rendering is equivalently decomposed into base-10 integer blocks as

$$\begin{aligned} \mathcal{I}_{512} &= 2953339382066878558096523387881 \times 10^{-33} \\ &\quad + 6292562558714204710746977471712 \times 10^{-64} \\ &\quad + 0232515199820836970136357290438 \times 10^{-95} \\ &\quad + 1962215104297226199323620401181 \times 10^{-126} \\ &\quad + 3775828345593770442070814599714 \times 10^{-157}. \end{aligned} \quad (28)$$

Consequently, the baseline tensor coefficients and off-diagonal constraints are

$$\begin{aligned} I_{00}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{11}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{22}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{33}(1 \text{ m}) &= \mathcal{I}_{512}, \\ \max_{\mu \neq \nu} |I_{\mu\nu}| &< 10^{-122}. \end{aligned} \quad (29)$$

The 512-bit significand provides a binary rounding unit and a decimal precision envelope of

$$\begin{aligned} u_{512} &= 2^{-513} = 3.729170365600103 \times 10^{-155}, \\ p_{10} &= 512 \log_{10} 2 = 154.1273578 \text{ digits}, \\ \epsilon_{\text{holo}} &= 1.0 \times 10^{-122}, \\ u_{512} &< \epsilon_{\text{holo}}. \end{aligned} \quad (30)$$

The compensated metric changes the ray deflection from the monopolar inverse-radius law to a linear aperture-radius law:

$$\begin{aligned} \hat{\alpha}_{\text{GR}}(r_0) &= \frac{2R_s}{r_0}, \\ \hat{\alpha}_{\text{eff}}(r_0) &= \frac{2R_s r_0}{r_{\text{ref}}^2}, \\ \mathcal{P}_{\text{eff}}(r_0) &= \frac{\hat{\alpha}_{\text{eff}}(r_0)}{r_0} = \frac{2R_s}{r_{\text{ref}}^2}, \\ \frac{d\mathcal{P}_{\text{eff}}}{dr_0} &= 0. \end{aligned} \quad (31)$$

Every paraxial ray within the commanded aperture therefore intersects a common plane:

$$\begin{aligned} f_{\text{planar}} &= \frac{r_0}{\hat{\alpha}_{\text{eff}}(r_0)} = \frac{r_{\text{ref}}^2}{2R_s}, \\ f_{\text{planar}} &= 169.2998789898903 \text{ m} \approx 169.30 \text{ m}. \end{aligned} \quad (32)$$

D. Electronic Beam Steering and Voltage Mapping

The 8×8 InP/InGaAs HBT array is indexed by $m, n \in \{0, \dots, 7\}$ and has pitch $d_e = 50.0 \mu\text{m}$. Coordinates relative to the electrical phase center are

$$\begin{aligned} x_m &= \left(m - \frac{7}{2}\right) d_e, \\ y_n &= \left(n - \frac{7}{2}\right) d_e, \\ d_e &= 50.0 \times 10^{-6} \text{ m}. \end{aligned} \quad (33)$$

For commanded wavefront tilts θ_x and θ_y , wavelength λ , and tensor-correction phase $\phi_{I,mn}$, each complex element command is

$$\begin{aligned} k_0 &= \frac{2\pi}{\lambda}, \\ \vartheta_{m,n} &= -k_0 (x_m \sin \theta_x + y_n \sin \theta_y) + \phi_{I,mn}, \\ a_{m,n} &= A_{m,n} \exp(i\vartheta_{m,n}), \\ \Theta_{\text{tilt}} &= \sqrt{\theta_x^2 + \theta_y^2}. \end{aligned} \quad (34)$$

The synthesized aperture factor and electronically translated beam maximum follow

$$\begin{aligned} \mathcal{A}(k_x, k_y) &= \sum_{m=0}^7 \sum_{n=0}^7 a_{m,n} \exp[i(k_x x_m + k_y y_n)], \\ k_{x,\max} &= k_0 \sin \theta_x, \\ k_{y,\max} &= k_0 \sin \theta_y. \end{aligned} \quad (35)$$

The phase command is wrapped into one 2π interval and mapped linearly between the HBT gate/base turn-on voltage and collector-drain saturation voltage:

$$\begin{aligned} \bar{\vartheta}_{m,n} &= \text{mod}(\vartheta_{m,n}, 2\pi), \\ V_{\text{on}} &= 3.800 \text{ V}, \\ V_{\text{sat}} &= 7.400 \text{ V}, \\ V_{m,n} &= V_{\text{on}} + \frac{V_{\text{sat}} - V_{\text{on}}}{2\pi} \bar{\vartheta}_{m,n}, \\ V_{m,n} &= 3.800 \text{ V} + (0.5729577951 \text{ V rad}^{-1}) \bar{\vartheta}_{m,n}. \end{aligned} \quad (36)$$

The 16-bit phase DAC code and the corresponding RF_PHASE_V register write at address 0x70000018 are

$$\begin{aligned} D_{m,n} &= \text{round} \left[65535 \frac{V_{m,n} - 3.800 \text{ V}}{3.600 \text{ V}} \right], \\ q &= 8m + n, \\ W_{m,n} &= (q \ll 16) \mid D_{m,n}, \\ 0 \leq q &\leq 63, \quad 0 \leq D_{m,n} \leq 65535, \\ \Delta V_{\text{LSB}} &= \frac{3.600 \text{ V}}{65535} = 54.9325 \mu\text{V}, \\ \Delta \vartheta_{\text{LSB}} &= \frac{2\pi}{65535} = 9.58753 \times 10^{-5} \text{ rad}. \end{aligned} \quad (37)$$

Closed-loop interpolation shall reduce the raw DAC quantization residual to the phase-stability allocation by dithering adjacent codes over the 72.0-GHz timing backplane:

$$\begin{aligned} f_{\text{clk}} &= 72.0 \text{ GHz}, \\ R_{\text{route}} &= 40.0 \text{ Gb s}^{-1} \text{ channel}^{-1}, \\ |\delta \vartheta_{m,n}| &\leq 5.05 \times 10^{-5} \text{ rad}. \end{aligned} \quad (38)$$

This command chain supplies non-mechanical two-axis retargeting while preserving the constant focal power of Eq. (31) and the 169.30-m planar detector interface of Eq. (32).

III. METRIC TRANSDUCER EMITTER ARRAY AND ACOUSTIC TAMPING ASSEMBLY

A. Heterojunction Bipolar Transducer Array Architecture

The metric transducer is a monolithic 8×8 InP/InGaAs heterojunction bipolar transistor (HBT) lattice. The channel coordinates, physical aperture, master timing, and deterministic state-routing capacity are

$$\begin{aligned} (m, n) &\in \{0, \dots, 7\}^2, \quad N_{\text{ch}} = 64, \\ d_e &= 50.0 \mu\text{m}, \\ L_{\text{ctr}} &= 7d_e = 350.0 \mu\text{m}, \\ f_{\text{master}} &= 72.0 \text{ GHz}, \\ R_{\text{bus}} &= 40.0 \text{ Gb s}^{-1} \text{ channel}^{-1}, \\ R_{\text{aggregate}} &= 64R_{\text{bus}} = 2.560 \text{ Tb s}^{-1}. \end{aligned} \quad (39)$$

The 72.0-GHz clock period and its relationship to the 19.82-MHz airbridge mode are

$$\begin{aligned} T_{\text{master}} &= \frac{1}{f_{\text{master}}} = 13.8889 \text{ ps}, \\ \frac{f_{\text{master}}}{f_{\text{bridge}}} &= \frac{72.0 \times 10^9}{19.82 \times 10^6} = 3.63269 \times 10^3. \end{aligned} \quad (40)$$

All mask vertices shall be quantized to the GDSII database grid

$$\Delta_{\text{GDS}} = 1.0 \text{ pm} = 1.0 \times 10^{-12} \text{ m}. \quad (41)$$

The controlling fabrication stack is specified in Table VI.

The Layer-20 airbridge is represented as a doubly clamped Euler-Bernoulli beam. Its unstressed and compression-shifted modal frequencies are

$$\begin{aligned} f_0 &= \frac{\beta_1^2 t}{2\pi \sqrt{12} L^2} \sqrt{\frac{E}{\rho}}, \\ \beta_1 &= 4.73004, \quad t = 1.50 \mu\text{m}, \quad L = 5.00 \mu\text{m}, \\ f_{\text{res}} &= f_0 \sqrt{1 - \frac{\sigma_0}{\sigma_{\text{crit}}}}, \\ \sigma_{\text{crit}} &= \frac{\pi^2 E t^2}{3L^2}. \end{aligned} \quad (42)$$

Using the niobium-dominated unstressed mode $f_0 = 215.88 \text{ MHz}$, the required compressive prestress ratio is

$$\begin{aligned} \frac{\sigma_0}{\sigma_{\text{crit}}} &= 1 - \left(\frac{f_{\text{res}}}{f_0} \right)^2 \\ &= 1 - \left(\frac{19.82}{215.88} \right)^2 = 0.991571. \end{aligned} \quad (43)$$

B. Closed-Loop PID Bias Regulation and Phase-Jitter Enforcement

Each HBT phase channel is regulated by a continuous-time proportional–integral–derivative compensator

$$\begin{aligned} C(s) &= K_p + \frac{K_i}{s} + K_d s \\ &= \frac{K_d s^2 + K_p s + K_i}{s}, \\ K_p &= 1.85 \text{ V rad}^{-1}, \\ K_i &= 9.12 \times 10^3 \text{ V rad}^{-1} \text{ s}^{-1}, \\ K_d &= 3.45 \times 10^{-7} \text{ V s rad}^{-1}. \end{aligned} \quad (44)$$

The compensator zeros are the roots of the numerator polynomial:

$$\begin{aligned} s_{1,2} &= \frac{-K_p \pm \sqrt{K_p^2 - 4K_d K_i}}{2K_d}, \\ s_1 &= -4.9342701 \times 10^3 \text{ rad s}^{-1}, \\ s_2 &= -5.3573846 \times 10^6 \text{ rad s}^{-1}, \\ f_1 &= \frac{|s_1|}{2\pi} = 785.3135 \text{ Hz}, \\ f_2 &= \frac{|s_2|}{2\pi} = 852.6542 \text{ kHz}. \end{aligned} \quad (45)$$

TABLE VI
METRIC-TRANSDUCER GDSII LITHOGRAPHIC MASK STACK

Layer	GDSII purpose	Material system	Thickness / geometry	Electrical and structural requirement
10	SUBSTRATE_INP	Semi-insulating InP (100)	350.0 μm wafer	RF-low-loss mechanical datum, cryogenic heat-spreading substrate, and HBT epitaxial support; all geometry resolved on the 1.0-pm database grid.
20	AIRBRIDGE_SPAN	Micromachined Au/InP bridge	1.5 $\mu\text{m} \times 5.0 \mu\text{m}$ span	Low-capacitance crossover with stress-shifted first flexural resonance $f_{\text{bridge}} = 19.82 \text{ MHz}$.
25	MET_NB_TRACE	Superconducting Nb	300.0 nm film	Zero-resistance bias routing below $T_c = 9.20 \text{ K}$ and direct coupling to the emergency current-shunt network.

For plant transfer function $P(s)$, the phase-error sensitivity and complementary sensitivity are

$$\begin{aligned} S(s) &= \frac{1}{1 + C(s)P(s)}, \\ T(s) &= \frac{C(s)P(s)}{1 + C(s)P(s)}, \\ S(s) + T(s) &= 1. \end{aligned} \quad (46)$$

Let $S_{\phi, \text{src}}(f)$ denote source phase-noise density and $S_{\phi, \text{dac}}(f)$ denote actuation and quantization noise. Hardware-in-the-loop acceptance requires

$$\begin{aligned} \sigma_\phi^2 &= \int_0^{f_{\text{Nyq}}} S_{\phi, \text{src}}(f) |S(j2\pi f)|^2 df \\ &\quad + \int_0^{f_{\text{Nyq}}} S_{\phi, \text{dac}}(f) |T(j2\pi f)|^2 df, \\ f_{\text{Nyq}} &= 36.0 \text{ GHz}, \\ \sigma_\phi^2 &\leq (5.05 \times 10^{-5} \text{ rad})^2 \\ &= 2.55025 \times 10^{-9} \text{ rad}^2. \end{aligned} \quad (47)$$

The deterministic peak monitor supplements the integrated-noise criterion. For all channels and every 13.8889-ps control frame, compliance is established by

$$\begin{aligned} \|\delta\phi[k]\|_\infty &= \max_{0 \leq q \leq 63} |\delta\phi_q[k]|, \\ \|\delta\phi[k]\|_\infty &\leq 5.05 \times 10^{-5} \text{ rad}, \\ \sigma_\phi &\leq 5.05 \times 10^{-5} \text{ rad}. \end{aligned} \quad (48)$$

Equations (47) and (48) jointly enforce both the RMS and samplewise HIL phase-jitter limits.

C. Sub-Nanosecond Acoustic Tamping Stack

The acoustic tamping path transfers the 142.08-MW field-quench transient from the sapphire waveguide through a quarter-wave nanoporous silica layer into the liquid-helium-4 bath:

$$\begin{aligned} P_{\text{quench, pk}} &= 142.08 \text{ MW}, \\ Z_1 &= 44.178 \text{ MRayl}, \\ Z_2 &= 0.0275 \text{ MRayl}, \\ T_{\text{bath}} &= 4.20 \text{ K}. \end{aligned} \quad (49)$$

At an unmediated sapphire-helium interface, the normal-incidence intensity reflection and transmission coefficients are

$$\begin{aligned} R_{\text{unmatched}} &= \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 \\ &= \left(\frac{44.178 - 0.0275}{44.178 + 0.0275} \right)^2 \\ &= 0.997513 = 99.7513\%, \\ \mathcal{T}_{\text{unmatched}} &= 1 - R_{\text{unmatched}} = 0.2487\%. \end{aligned} \quad (50)$$

The ideal geometric-mean impedance is 1.102223 MRayl; the flight layer is deliberately offset to 1.1512 MRayl to include cryogenic densification and bonded-interface corrections:

$$\begin{aligned} Z_{\text{geom}} &= \sqrt{Z_1 Z_2} = 1.102223 \text{ MRayl}, \\ Z_m &= 1.1512 \text{ MRayl}, \\ d_m &= 6.395 \text{ nm}, \\ d_m &= \frac{\lambda_m}{4} = \frac{v_m}{4f_{\text{ac}}}. \end{aligned} \quad (51)$$

At the quarter-wave design frequency, the transformed load and matched reflection coefficient are

$$\begin{aligned} Z_{\text{in}} &= \frac{Z_m^2}{Z_2} = 48.1913 \text{ MRayl}, \\ R_{\text{matched}} &= \left(\frac{Z_1 - Z_{\text{in}}}{Z_1 + Z_{\text{in}}} \right)^2 \\ &= 1.888 \times 10^{-3} = 0.1888\%, \\ \mathcal{T}_{\text{matched}} &= 1 - R_{\text{matched}} = 99.8112\%. \end{aligned} \quad (52)$$

The corresponding peak reflected and transmitted powers are

$$\begin{aligned} P_{\text{refl, pk}} &= R_{\text{matched}} P_{\text{quench, pk}} = 268.26 \text{ kW}, \\ P_{\text{trans, pk}} &= \mathcal{T}_{\text{matched}} P_{\text{quench, pk}} = 141.812 \text{ MW}. \end{aligned} \quad (53)$$

The transient pressure acceptance values and material margins are summarized in Table VII.

The pressure attenuation and governing InP safety factor are

$$\begin{aligned} \mathcal{A}_P &= 1 - \frac{P_{\text{InP}}}{P_{\text{sapphire}}} = 1 - \frac{8.68}{12.64} = 31.329\%, \\ \Delta P_{\text{margin}} &= Y_{\text{InP}} - P_{\text{InP}} \\ &= 10.00 \text{ GPa} - 8.68 \text{ GPa} = 1.32 \text{ GPa}, \\ \mathcal{M}_Y &= \frac{\Delta P_{\text{margin}}}{Y_{\text{InP}}} = 13.20\%. \end{aligned} \quad (54)$$

TABLE VII
SUB-NANOSECOND ACOUSTIC PRESSURE TRANSMISSION AND YIELD MARGINS

Interface	E (GPa)	P_{pk} (GPa)	Y (GPa)	Margin (GPa)	Disposition
Single-crystal sapphire	400.00	12.64	22.00	9.36	Input waveguide remains below the compressive acceptance limit.
Nanoporous silica aerogel	0.05	0.31	0.85	0.54	Quarter-wave layer remains within the bonded porous-medium allowables.
Semi-insulating InP interface	61.00	8.68	10.00	1.32	Governing interface; positive 13.20% yield margin.

D. Thermomechanical Stress and Coffin–Manson Fatigue

A three-dimensional Chaboche model resolves cyclic thermoviscoplastic strain across the alumina/InP/Nb interfaces. Total strain is decomposed according to

$$\begin{aligned}\boldsymbol{\varepsilon} &= \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p + \boldsymbol{\varepsilon}^{th}, \\ \boldsymbol{\sigma} &= \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p - \boldsymbol{\varepsilon}^{th}), \\ \boldsymbol{\varepsilon}^{th}(T) &= \int_{T_0}^T \alpha_T(\vartheta) d\vartheta \mathbf{1}, \\ \Delta T &= 15.0 \text{ K}.\end{aligned}\quad (55)$$

The thermal-expansion mismatch inputs are

$$\begin{aligned}\alpha_{\text{Al}_2\text{O}_3} &= 5.4 \times 10^{-6} \text{ K}^{-1}, \\ \alpha_{\text{InP}} &= 4.6 \times 10^{-6} \text{ K}^{-1}, \\ \alpha_{\text{Nb}} &= 7.3 \times 10^{-6} \text{ K}^{-1}.\end{aligned}\quad (56)$$

With deviatoric stress $\boldsymbol{s} = \text{dev}(\boldsymbol{\sigma})$, two kinematic backstress tensors $\boldsymbol{X}_j$, and isotropic hardening R , the yield surface and viscoplastic flow rule are

$$\begin{aligned}\boldsymbol{\xi} &= \boldsymbol{s} - \sum_{j=1}^2 \boldsymbol{X}_j, \\ f &= \sqrt{\frac{3}{2} \boldsymbol{\xi} : \boldsymbol{\xi} - (\sigma_y + R)}, \\ \dot{p} &= \left\langle \frac{f}{K} \right\rangle^n, \\ \dot{\boldsymbol{\varepsilon}}^p &= \dot{p} \frac{3}{2} \frac{\boldsymbol{\xi}}{\sqrt{\frac{3}{2} \boldsymbol{\xi} : \boldsymbol{\xi}}}.\end{aligned}\quad (57)$$

The Chaboche internal variables evolve as

$$\begin{aligned}\dot{\boldsymbol{X}}_j &= \frac{2}{3} C_j \dot{\boldsymbol{\varepsilon}}^p - \gamma_j \boldsymbol{X}_j \dot{p} + \frac{\partial \boldsymbol{X}_j}{\partial T} \dot{T}, \quad j \in \{1, 2\}, \\ \dot{R} &= b(Q - R) \dot{p}, \\ \dot{p} &= \sqrt{\frac{2}{3} \dot{\boldsymbol{\varepsilon}}^p : \dot{\boldsymbol{\varepsilon}}^p}.\end{aligned}\quad (58)$$

Low-cycle fatigue is closed with the Coffin–Manson relation. For a calibrated fatigue ductility exponent $c = -0.5000$, the specified strain range and cycle life imply

$$\begin{aligned}\frac{\Delta \varepsilon_p}{2} &= \varepsilon'_f (2N_f)^c, \\ \Delta \varepsilon_p &= 6.0 \times 10^{-6}, \\ N_f &= 4.0 \times 10^6 \text{ cycles}, \\ \varepsilon'_f &= \frac{\Delta \varepsilon_p}{2} (2N_f)^{-c} = 8.4853 \times 10^{-3}.\end{aligned}\quad (59)$$

The mechanical cycle life maps to the de-rendering allocation as

$$\begin{aligned}B_{\text{life}} &= 1.514 \times 10^{16} \text{ bit}, \\ b_{\text{cycle}} &= \frac{B_{\text{life}}}{N_f} \\ &= \frac{1.514 \times 10^{16}}{4.0 \times 10^6} = 3.7850 \times 10^9 \text{ bit cycle}^{-1}.\end{aligned}\quad (60)$$

E. Fatigued Impedance and Post-Shunt Quench Margin

At the fatigue limit, aerogel densification increases the matching-layer impedance to 1.3250 MRayl. The resulting quarter-wave input impedance and reflected-power fraction are

$$\begin{aligned}Z_{\text{fatigue}} &= 1.3250 \text{ MRayl}, \\ Z_{\text{in,fatigue}} &= \frac{Z_{\text{fatigue}}^2}{Z_2} = 63.8909 \text{ MRayl}, \\ R_{\text{fatigue}} &= \left(\frac{Z_1 - Z_{\text{in,fatigue}}}{Z_1 + Z_{\text{in,fatigue}}} \right)^2 \\ &= 0.033136 = 3.3136\%.\end{aligned}\quad (61)$$

Relative to the initially transmitted quench power, the reflected-power escalation is

$$\begin{aligned}\Delta P_{\text{refl,rel}} &= \frac{R_{\text{fatigue}} - R_{\text{matched}}}{1 - R_{\text{matched}}} \\ &= \frac{0.033136 - 0.001888}{1 - 0.001888} \\ &= 0.031307 = 3.13\%.\end{aligned}\quad (62)$$

The InP volumetric heat capacity follows the low-temperature Debye law

$$\begin{aligned}C_{v,\text{vol}}(T) &= a_{\text{InP}} T^3, \\ a_{\text{InP}} &= 3.8776 \text{ J m}^{-3} \text{ K}^{-4}.\end{aligned}\quad (63)$$

The energy density required to heat the substrate from the helium-bath temperature to the niobium quench acceptance temperature is

$$\begin{aligned}Q_{\text{quench}} &= \int_{T_{\text{bath}}}^{T_{\text{quench}}} a_{\text{InP}} T^3 dT \\ &= \frac{a_{\text{InP}}}{4} (T_{\text{quench}}^4 - T_{\text{bath}}^4), \\ T_{\text{bath}} &= 4.2000 \text{ K}, \\ T_{\text{quench}} &= 9.3000 \text{ K}, \\ Q_{\text{quench}} &= 6949.97 \text{ J m}^{-3}.\end{aligned}\quad (64)$$

TABLE VIII
CRYOGENIC QUENCH-STATE MARGINS

State	T (K)	Headroom (K)
He-4 base bath	4.2000	5.1000
Post-shunt peak	6.8638	2.4362
Nb quench acceptance	9.3000	0.0000

For a 2.50-ns current-shunt deposition of $Q_{\text{shunt}} = 1850.00 \text{ J m}^{-3}$, inversion of the Debye integral gives

$$\begin{aligned} T_{\text{peak}} &= \left(T_{\text{bath}}^4 + \frac{4Q_{\text{shunt}}}{a_{\text{InP}}} \right)^{1/4} \\ &= \left[(4.2000)^4 + \frac{4(1850.00)}{3.8776} \right]^{1/4} \text{ K} \\ &= 6.8638 \text{ K}. \end{aligned} \quad (65)$$

The positive niobium quench headroom is therefore

$$\begin{aligned} \Delta T_{\text{headroom}} &= T_{\text{quench}} - T_{\text{peak}} \\ &= 9.3000 \text{ K} - 6.8638 \text{ K} \\ &= 2.4362 \text{ K} > 0. \end{aligned} \quad (66)$$

Table VIII closes the transient thermal acceptance chain.

IV. MODULAR LANR POWER INFRASTRUCTURE & DERATING PROTOCOLS

A. $N + 14$ Redundant LANR Power Topology & Bus Management

The installed LANR power plant comprises $N_{\text{total}} = 1800$ independently isolatable modules connected through segmented, cross-strapped dc buses. With the verified module output $P_m = 507.3195 \text{ W}$, the nominal net plant output and reserve above the continuous entropy-debt load are

$$\begin{aligned} P_{\text{net}, N+14} &= N_{\text{total}} P_m \\ &= (1800)(507.3195 \text{ W}) = 913.1751 \text{ kW}, \\ \Delta P_{N+14} &= P_{\text{net}, N+14} - P_{\text{debt}} \\ &= 913.1751 \text{ kW} - 906.0000 \text{ kW} \\ &= +7.1751 \text{ kW}, \\ \mathcal{M}_P &= 100 \frac{\Delta P_{N+14}}{P_{\text{debt}}} = +0.7920\%. \end{aligned} \quad (67)$$

The absolute module floor that sustains the full-power seed is

$$\begin{aligned} N_{\text{min}} &= \left\lceil \frac{906000.0000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil \\ &= 1786, \quad N_{\text{total}} = N_{\text{min}} + 14. \end{aligned} \quad (68)$$

The bus-management controller isolates a failed module, closes the appropriate cross-strap, and rebalances channel current without interrupting the non-sheddable metric load. Table IX defines the operational states from the installed $N + 14$ topology through the $N + 0$ absolute minimum. Every state at or above $N + 0$ maintains $M_{\text{seed}} = 10^{-6} M_{\odot}$ and $f_0 = 169.30 \text{ m}$; outages from $N + 13$ through $N + 1$ therefore require neither seed-mass derating nor Sensor Craft repositioning.

B. Entropy-Debt Power Requirement

The power infrastructure shall continuously compensate the entropy debt associated with the commanded ghost-seed mass. The governing proportional law is

$$\begin{aligned} P_{\text{debt}}(M_{\text{seed}}) &= \left(\frac{M_{\text{seed}}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}), \\ P_{\text{debt}}(10^{-6} M_{\odot}) &= 906.00 \text{ kW}. \end{aligned} \quad (69)$$

This load is treated as continuous and non-sheddable while the metric perturbation is active. The generation bus shall therefore provide at least 906 000 W net after all module-local pumps, chillers, controls, and conversion losses have been charged against gross production.

C. Modular LANR Cold-Fusion Cluster

Each verified SHBT LANR module combines a 2911.40-W fusion source with 182.04 W of absorbed optical power. Its gross thermal balance is

$$\begin{aligned} P_{\text{thermal}} &= P_{\text{fusion}} + P_{\text{opt,abs}} \\ &= 2911.40 \text{ W} + 182.04 \text{ W} \\ &= 3093.44 \text{ W}. \end{aligned} \quad (70)$$

A dual-stage skutterudite/half-Heusler thermoelectric generator (TEG) converts this heat at a specified efficiency of $\eta_{\text{TEG}} = 0.3380$:

$$\begin{aligned} P_{\text{TEG}, \text{elec}} &= \eta_{\text{TEG}} P_{\text{thermal}} \\ &= (0.3380)(3093.44 \text{ W}) \\ &= 1045.58 \text{ W}. \end{aligned} \quad (71)$$

The complete parasitic ledger is $P_{\text{house}} = 538.26 \text{ W}$ per module (538.2632 W before display rounding), including the 27.24-W primary circulation pump and the 58.30-W auxiliary chiller. The remaining 452.72 W covers reactor control, secondary circulation, power conditioning, instrumentation, and module thermal management. Consequently,

$$\begin{aligned} P_{\text{net}, \text{complete}} &= P_{\text{TEG}, \text{elec}} - P_{\text{house}} \\ &= 1045.5827 \text{ W} - 538.2632 \text{ W} \\ &= +507.3195 \text{ W module}^{-1}. \end{aligned} \quad (72)$$

The minimum full-seed count and installed-cluster output are therefore

$$\begin{aligned} N_{\text{min}} &= \left\lceil \frac{P_{\text{debt}}}{P_{\text{net}, \text{complete}}} \right\rceil \\ &= \left\lceil \frac{906000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil = 1786, \\ P_{\text{cluster}, \text{net}} &= (1800)(507.3195 \text{ W}) \\ &= 913.1751 \text{ kW}. \end{aligned} \quad (73)$$

The installed topology provides 14 module-out capability above the absolute minimum. Seed derating and focal repositioning begin only after the active population falls below $N_{\text{min}} = 1786$, as specified in Section VI-E.

TABLE IX
OPERATIONAL LANR REDUNDANCY STATES

State	Active modules	Redundant modules	Net power (kW)	Reserve (kW)	Bus-management disposition
$N + 14$ nominal	1800	14	913.1751	+7.1751	Baseline load distribution across all available channels.
$N + 13$ single outage	1799	13	912.6678	+6.6678	Failed module isolated; optical-drive allocation unchanged.
$N + 12$ double outage	1798	12	912.1605	+6.1605	Remaining channels absorb the load through current rebalancing.
$N + 10$ degraded	1796	10	911.1458	+5.1458	Secondary diagnostics are power-limited; metric synthesis remains fully powered.
$N + 5$ intermediate	1791	5	908.6092	+2.6092	Optical phase synthesis is prioritized over nonessential loads.
$N + 1$ critical margin	1787	1	906.5799	+0.5799	Critical-state telemetry is asserted and discretionary loads are shed.
$N + 0$ absolute minimum	1786	0	906.0726	+0.0726	Full seed mass and the 169.30-m focal baseline are maintained at the redundancy boundary.

TABLE X
RADIATOR AREA VERSUS OPERATING TEMPERATURE

T_{rad} (K)	A_{radiator} (m ²)
400	3485.63
450	2176.06
500	1427.72
550	975.15
600	688.52
650	499.88
700	371.65

D. Deep-Space Radiator and Thermal Balance

For the complete cluster, the gross heat production and net exported electrical power give

$$\begin{aligned}
 P_{\text{cluster,thermal}} &= (1800)(3093.44 \text{ W}) = 5.568192 \text{ MW}, \\
 P_{\text{waste}} &= P_{\text{cluster,thermal}} - P_{\text{cluster,net}} \\
 &= 4.6550169 \text{ MW} \simeq 4.6550 \text{ MW}.
 \end{aligned} \tag{74}$$

The required effective radiating area follows the Stefan–Boltzmann balance

$$A_{\text{radiator}}(T_{\text{rad}}) = \frac{P_{\text{waste}}}{\epsilon \sigma (T_{\text{rad}}^4 - T_{\text{space}}^4)}, \tag{75}$$

where $\epsilon = 0.92000$, $T_{\text{space}} = 2.7 \text{ K}$, and $\sigma = 5.670374419 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Table X shows the strong fourth-power sensitivity of deployed area to radiator temperature.

At the 600-K baseline, Eq. (75) requires $A_{\text{radiator}} = 688.52 \text{ m}^2$ of effective emitting surface for 4.6550 MW of continuous waste-heat rejection. Packaging shall distinguish effective emitting area from projected planform area when both faces of a radiator panel have unobstructed views to deep space.

E. Active Entropic Refrigeration

The 4.2-K boundary assembly uses active entropic refrigeration to remove the heat associated with controlled de-rendering events. If Γ_{de} is the event rate and each event removes one

bit with entropy increment $\Delta S = k_B \ln 2$, the ideal cold-side cooling rate is

$$\begin{aligned}
 P_{\text{cool}} &= \Gamma_{\text{de}} \Delta S T_c \\
 &= \Gamma_{\text{de}} k_B T_c \ln 2, \\
 T_c &= 4.2 \text{ K}, \quad k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}.
 \end{aligned} \tag{76}$$

Thus, the ideal event rate required per watt of cold-side removal is

$$\frac{\Gamma_{\text{de}}}{P_{\text{cool}}} = \frac{1}{k_B T_c \ln 2} = 2.4880 \times 10^{22} \text{ bit J}^{-1}. \tag{77}$$

Let $q_b'' = P_b/A_b$ denote the net heat flux crossing boundary area A_b , and let ΔT_b be the corresponding temperature rise. The refrigeration controller shall command Γ_{de} such that

$$\begin{aligned}
 P_{\text{cool}} &\geq P_b, \\
 R_{\text{th},b}'' &= \frac{\Delta T_b}{q_b''} < 7.04 \times 10^{-6} \text{ m}^2 \text{ K W}^{-1}.
 \end{aligned} \tag{78}$$

The control law shall use the measured boundary heat load, not the ideal entropy term alone, when setting the commanded event rate; conversion inefficiency and parasitic cryogenic loads are carried as additional cold-stage margin.

V. C-ABI ZERO-COPY SHARED MEMORY TELEMETRY ARCHITECTURE

A. POSIX Shared-Memory SPSC Transport

High-rate telemetry is transported through two unidirectional POSIX shared memory rings: a driver-to-shbt-os acquisition ring and a shbt-os-to-user-daemon publication ring. Each object is created with `shm_open`, sized once with `ftruncate`, and mapped into the participating address spaces with `mmap(..., MAP_SHARED, ...)`. The hardware driver maps DMA destination pages directly onto producer-owned frame slots, and the consumer reads the committed slot in place. No payload copy, serialization pass, system-call transition, or heap allocation occurs in the steady-state data path.

Each ring is a power-of-two array of Q telemetry frames with monotonic producer and consumer indices

$$\begin{aligned} h &= \text{write_head}, & t &= \text{read_tail}, \\ n_{\text{used}} &= h - t, & q_{\text{quad}} n_{\text{free}} &= Q - (h - t), \\ i_{\text{write}} &= h \& (Q - 1), & q_{\text{quad}} i_{\text{read}} &= t \& (Q - 1). \end{aligned} \quad (79)$$

The producer owns `write_head`; the consumer owns `read_tail`. Both are C-ABI-compatible `AtomicUsize` values. The producer writes a complete frame and publishes $h + 1$ with release ordering. The consumer loads h with acquire ordering, validates the frame, processes it in place, and publishes $t + 1$ with release ordering. Full and empty states are defined by $h - t = Q$ and $h = t$, respectively. Unsigned monotonic-index subtraction makes wraparound well defined for ring occupancies smaller than half the index space.

The frame type and both index records use strict cache-line layout equivalent to `#[repr(C, align(64))]`. The write-head record, read-tail record, and frame array begin on separate 64-byte cache lines. Consequently, the producer never writes the consumer's cache line and the consumer never writes the producer's cache line, eliminating false sharing and cache-line bouncing from the SPSC synchronization path.

The frame-size and alignment invariants are

$$\begin{aligned} \text{sizeof}(\text{TelemetryFrame}) &= 64 \text{ byte}, \\ \text{alignof}(\text{TelemetryFrame}) &= 64 \text{ byte}, \\ \text{offsetof}(\text{crc16}) &= 62 \text{ byte}. \end{aligned} \quad (80)$$

All multibyte fields use little-endian representation. A frame becomes visible only after DMA completion, reserved-byte zeroing, sequence-number assignment, status capture, and CRC generation. A CRC failure or discontinuous `sequence_id` marks the slot invalid without exposing partial telemetry to the deterministic control state.

B. Bare-Metal Microkernel and SRAM Arena

The flight-control computer shall execute `shbt-os` as a freestanding, bare-metal C11 environment. The build shall use the freestanding implementation model, provide no heap allocator, perform no dynamic allocation after reset, and link against no hosted C library. Startup code, fixed-width arithmetic, memory primitives, interrupt entry, and MMIO access shall be supplied by the verified microkernel image. All control paths shall have statically bounded memory use and shall avoid recursion, unbounded loops, and data-dependent allocation.

The linker shall reserve one contiguous `UnifiedStinespringFrame` in on-chip SRAM, align its base to a 64-byte boundary, and assert its size at link time:

$$\begin{aligned} A_{\text{frame}} \bmod 64 &= 0, \\ B_{\text{frame}} &= 2112 \text{ byte}, \\ B_A &= 640 \text{ byte} = \frac{10}{33} B_{\text{frame}}, \\ B_D &= 1472 \text{ byte} = \frac{23}{33} B_{\text{frame}}, \\ B_A + B_D &= 2112 \text{ byte}. \end{aligned} \quad (81)$$

The Active Visible Register occupies the first 640 bytes and has rational dilation weight $\eta_A = 10/33$. It contains the live

64-channel state vector, threshold buffers, command shadow registers, and current-cycle ECC work products. The Dark Ledger occupies the following 1472 bytes with $\eta_D = 23/33$. Of this region, 124 fixed-width braid descriptors consume

$$B_{\text{braid}} = (124)(8 \text{ byte}) = 992 \text{ byte}, \quad (82)$$

leaving 480 bytes for ledger indices, ECC syndromes, recovery checkpoints, and deterministic padding. The identities $\eta_A + \eta_D = 1$ and $B_A : B_D = 10 : 23$ shall be compile-time invariants.

C. Normative SHBT-MMIO-1 Core Register Map

Table XII is the normative SHBT-MMIO-1 core register block. Fourteen contiguous 32-bit registers span byte offsets 0x00 through 0x34; hence the occupied interval is

$$B_{\text{MMIO}} = 0x38 = 56 \text{ byte}, \quad A_{\text{base}} = 0x70000000. \quad (83)$$

All registers are naturally aligned, strongly ordered, and excluded from caches. The interference-tensor aperture beginning at 0x70000040 is an extension window and is not included in the 56-byte core span.

D. SECDED ECC and Constant-Time Channel Remapping

Every persistent 64-bit SRAM payload shall be stored as a Hamming (72,64) SECDED codeword. Seven Hamming check bits occupy the power-of-two positions in the first 71 codeword locations, and the seventy-second bit supplies overall parity. For received codeword c , the decoder forms the seven-bit syndrome s and overall-parity mismatch p and applies

$$\begin{aligned} s = 0, p = 0 &\Rightarrow \text{no error}, \\ s \neq 0, p = 1 &\Rightarrow \text{correct one indexed bit}, \\ s = 0, p = 1 &\Rightarrow \text{correct the overall-parity bit}, \\ s \neq 0, p = 0 &\Rightarrow \text{detect double-bit error}. \end{aligned} \quad (84)$$

Correction shall occur before a payload enters the Active Visible Register. A double-bit detection shall inhibit the affected command, latch `ECC_SYNDROME`, and invoke channel replacement rather than release corrupted state to the transducer array.

Dynamic replacement is performed by `shbt_remap`, which applies an AVX-512 Givens rotation to the failed and spare channel components. For channel indices i and j ,

$$\begin{bmatrix} x'_i \\ x'_j \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix}. \quad (85)$$

The selected coefficient pair is loaded from a fixed table, and masked permutes replace the selected lanes without scanning the channel set. One vector load, one fixed shuffle/multiply-add schedule, and one vector store therefore give $O(1)$ execution with respect to the number of active channels. The routine shall preserve the vector norm to the fixed-point rounding bound and shall commit `REMAP_SRC` and `REMAP_DST` before the replacement channel is enabled.

TABLE XI
COMPLETE 64-BYTE CACHE-ALIGNED TELEMETRY FRAME LAYOUT

Byte offset	Field	C type	Representation	C-ABI function and reserved-byte rule
0x00–0x07	timestamp	uint64_t	Epoch ns	Unsigned nanoseconds from the configured mission epoch.
0x08–0x27	phase_error[13]	int16_t[13]	Q15 rad	Thirteen signed phase residuals occupy 0x08–0x21; 0x22–0x27 are zeroed reserved padding. Physical phase is $q/2^{15}$ rad.
0x28–0x2F	substrate_temp[1]	uint16_t[1]	Q8.8 K	The temperature sample occupies 0x28–0x29; 0x2A–0x2F are zeroed reserved padding. Physical temperature is $q/2^8$ K.
0x30–0x37	piezo_stress[1]	uint16_t[1]	Q8.8 MPa	The stress sample occupies 0x30–0x31; 0x32–0x37 are zeroed reserved padding. Physical stress is $q/2^8$ MPa.
0x38–0x3B	sequence_id	uint32_t	Unsigned counter	Monotonic frame identifier for loss, duplication, and ordering detection.
0x3C–0x3D	status_flags	uint16_t	Bit field	Encodes the $N + k$ power state, wide-angle steering state, metrology validity, ECC state, and shunt status.
0x3E–0x3F	crc16	uint16_t	CCITT-16	CRC-16/CCITT over bytes 0x00–0x3D; the receiver validates it before advancing read_tail.

TABLE XII
NORMATIVE SHBT-MMIO-1 CORE REGISTER MAP

Address	Symbol	Access	Width	Normative function
0x70000000	CR0_CTRL	R/W	32 bit	Core enable, operating-mode selection, phase-loop arm, and synchronous soft reset.
0x70000004	SR0_STAT	R	32 bit	Phase-lock, thermal-alarm, metrology-valid, ECC, quench, and shunt-latched status.
0x70000008	DET_MU_HI	R	32 bit	Upper word of signed Q1.62 eigenvector-detuning telemetry.
0x7000000C	DET_MU_LO	R	32 bit	Lower word of signed Q1.62 eigenvector-detuning telemetry.
0x70000010	BIT_OVF_N	R/W	32 bit	Upper command word of the 64-bit boundary-state overflow mantissa.
0x70000014	BIT_OVF_L	R/W	32 bit	Lower command word of the 64-bit boundary-state overflow mantissa.
0x70000018	RF_PHASE_V	R/W	32 bit	Channel address and phase-bias DAC command.
0x7000001C	SHUNT_TRIG	W	32 bit	Any nonzero write asserts the emitter-current diversion network; a zero write has no effect.
0x70000020	REC_SEQ_PC	R/W	32 bit	Write loads the recovery entry point and starts execution; read returns the microsequence PC and terminal code.
0x70000024	ECC_SYNDROME	R	32 bit	Latest SECDED syndrome, corrected-bit index, and single-/double-error classification.
0x70000028	ECC_CORR_ADDR	R	32 bit	SRAM word address associated with the most recent corrected or uncorrectable codeword.
0x7000002C	REMAP_SRC	R/W	32 bit	Failed logical-channel identifier and remapping generation number.
0x70000030	REMAP_DST	R/W	32 bit	Replacement physical-channel identifier and Givens-rotation slot.
0x70000034	IRQ_MASK_ACK	R/W	32 bit	Interrupt mask on write and latched-source acknowledgement on write-one-to-clear.

E. AVX-512 SIMD Safety-Interlock Pipeline

The highest-criticality monitor operates on two 64-byte stack buffers whose addresses satisfy $A \bmod 64 = 0$. One holds sixteen lane-wise detuning magnitudes and the other holds the replicated 10^{-12} threshold. The normative comparison pipeline is

$$\text{vmovaps} \rightarrow \text{vcmppps} \rightarrow \text{vmovmskps} \rightarrow \text{mov}, \quad (86)$$

where the compare is true for any lane satisfying $|\delta\mu_i| > 10^{-12}$. The resulting nonzero mask is written directly to SHUNT_TRIG; because a zero write is defined as a no-operation, the MMIO assertion path contains no conditional branch. Four vector groups cover all 64 channels through a fixed unrolled schedule and feed a hardware OR reduction before the single MMIO write.

At a 4.000-GHz safety clock, the scheduled load, compare, mask extraction, reduction, and strongly ordered write issue

in six cycles:

$$\begin{aligned} t_{\text{pipe}} &= \frac{6}{4.000 \times 10^9 \text{ s}^{-1}} = 1.50 \text{ ns}, \\ t_{\text{window}} &= 2.50 \text{ ns}, \\ t_{\text{margin}} &= t_{\text{window}} - t_{\text{pipe}} = 1.00 \text{ ns}. \end{aligned} \quad (87)$$

The MMIO bridge shall provide a deterministic bypass path such that a nonzero mask reaches 0x7000001C and asserts the current shunt before 2.50 ns from the sampled threshold violation. The governing predicate is

$$\max_i |\delta\mu_i| > 10^{-12} \implies \text{SHUNT_TRIG} \leftarrow 0xFFFFFFFF. \quad (88)$$

F. Thermal Hold and Deterministic Recovery

Shunt assertion begins the thermal stabilization interval at $t = 2.50$ ns. Hardware holds all phase-bias outputs below turn-on while the acoustic and niobium transients decay; the hold remains active through $t = 110$ ns. Once thermal-valid, phase-valid, and ECC-valid are simultaneously asserted,

shbt-os writes the fixed shbt_recover entry point to REC_SEQ_PC at 0x70000020. The sequencer restores the last ECC-clean checkpoint, applies any pending channel remap, and re-enables the phase loop according to

$$\begin{aligned} t_{\text{shunt}} &\leq 2.50 \text{ ns}, \\ 2.50 \text{ ns} &\leq t_{\text{thermal}} \leq 110 \text{ ns}, \\ t_{\text{recover}} &\leq 120.0 \text{ ns}, \\ t_{\text{recover}} - 110 \text{ ns} &\leq 10.0 \text{ ns}. \end{aligned} \quad (89)$$

Any failed validity check, uncorrectable SECDED event, or renewed rigidity violation leaves the shunt latched and prevents the recovery PC from advancing to the phase-enable terminal state.

VI. IN-FLIGHT GNC & 10-YEAR PROPELLANT LIFECYCLE ACCOUNTING

A. Dual-Wavelength Heterodyne Laser Interferometry and DWS

The inter-craft metrology system combines two frequency-stabilized optical carriers,

$$\lambda_1 = 1064.500 \text{ nm}, \quad \lambda_2 = 1064.520 \text{ nm}, \quad (90)$$

to provide optical-fringe precision together with a synthetic wavelength for integer-fringe resolution. The resulting synthetic wavelength is

$$\lambda_{\text{syn}} = \frac{\lambda_1 \lambda_2}{|\lambda_1 - \lambda_2|} = 56.6583 \text{ mm}. \quad (91)$$

Balanced InP/InGaAs photodetectors reject common-mode intensity noise and downconvert the received optical fields to an 80.0-MHz intermediate-frequency beat. A digital phase-locked loop unwraps the synthetic-wavelength phase and then estimates fine range from the carrier phase,

$$\Phi_{\text{het}}(t) = \frac{4\pi}{\lambda_1} r(t) + \phi_{\text{inst}}(t) + \phi_{\text{noise}}(t). \quad (92)$$

For $f > 1.0$ Hz, the one-sided phase-noise amplitude spectral density shall satisfy

$$S_{\phi}^{1/2}(f) \leq 2.0 \text{ } \mu\text{rad}/\sqrt{\text{Hz}}, \quad (93)$$

which corresponds to a single-axis range-noise amplitude spectral density

$$S_r^{1/2}(f) = \frac{\lambda_1}{4\pi} S_{\phi}^{1/2}(f) \leq 0.1694 \text{ pm}/\sqrt{\text{Hz}}. \quad (94)$$

In a 1-Hz equivalent noise bandwidth this gives $\sigma_r \leq 0.17$ pm. Bias calibration, cyclic nonlinearity, clock transfer, beam walk, and thermoelastic drift are separately estimated in the state filter; their root-sum-square allocation shall preserve the formation requirement $\|\delta \mathbf{r}\|_{3\sigma} \leq 1.00$ nm over the 169.30-m reference baseline.

Angular knowledge is obtained by differential wavefront sensing (DWS) on redundant quadrant-photodiode (QPD) arrays. Opposed-quadrant heterodyne phases form pitch and yaw observables,

$$e_{\theta,x} = K_{\text{DWS}}^{-1}(\bar{\Phi}_A - \bar{\Phi}_B), \quad e_{\theta,y} = K_{\text{DWS}}^{-1}(\bar{\Phi}_C - \bar{\Phi}_D), \quad (95)$$

where K_{DWS} is calibrated against beam size, curvature, and QPD geometry. The fused attitude solution shall maintain $\sigma_{\theta} \leq 0.015 \text{ } \mu\text{rad} = 15 \text{ nrad}$ per axis, with sub-nanoradian readout resolution retained for control-law quantization and calibration.

B. Sun–Earth L2 Environmental Disturbance Modeling

Solar-radiation pressure (SRP) on either craft is modeled by

$$\mathbf{a}_{\text{SRP}} = -\frac{C_r S_0}{c} \frac{A}{m} \left(\frac{1 \text{ AU}}{R} \right)^2 \hat{\mathbf{s}}, \quad (96)$$

where $S_0 = 1361 \text{ W m}^{-2}$, C_r is the effective radiation-pressure coefficient, and $\hat{\mathbf{s}}$ points toward the Sun. The Lens Craft and Sensor/Focal Craft area-to-mass ratios, approximately 0.0150 and 0.0042 $\text{m}^2 \text{ kg}^{-1}$, respectively, produce the modeled differential acceleration

$$\Delta \mathbf{a}_{\text{SRP}} \approx 3.42 \times 10^{-8} \text{ m s}^{-2} \hat{\mathbf{s}}. \quad (97)$$

The linearized Sun–Earth gravity-gradient contribution is propagated with the time-varying tidal tensor evaluated along the halo-orbit reference trajectory,

$$\begin{aligned} \mathbf{a}_{\text{grav,diff}}(t) &= \mathbf{C}_{\text{tidal}}(t) \delta \mathbf{r}(t), \\ \mathbf{C}_{\text{tidal}} &= \sum_{j \in \{\odot, \oplus\}} \frac{GM_j}{R_j^3} \left(3\hat{\mathbf{R}}_j \hat{\mathbf{R}}_j^T - \mathbf{I} \right). \end{aligned} \quad (98)$$

At the nominal L2 geometry, the combined Sun–Earth tensor produces an axial differential acceleration of order $4.7 \times 10^{-11} \text{ m s}^{-2}$ over 169.30 m; the flight propagator shall use ephemeris-resolved $\mathbf{C}_{\text{tidal}}(t)$ rather than this scalar design value.

Residual outgassing is represented by a decaying bias plus stochastic acceleration,

$$\mathbf{a}_{\text{out}}(t) = \mathbf{a}_{0,\text{out}} e^{-t/\tau_{\text{out}}} + \sqrt{2D_{\text{out}}} \xi(t), \quad \tau_{\text{out}} = 14.0 \text{ day}, \quad (99)$$

with $S_{a,\text{out}}^{1/2}(f) \leq 1.0 \times 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$. Thermoelastic metrology bias follows

$$\Delta L(t) = \alpha_{\text{CTE}} L_b \Delta T(t), \quad \alpha_{\text{CTE}} = 1.2 \times 10^{-7} \text{ K}^{-1}, \quad (100)$$

where the $L_b = 3.75$ -mm effective C/SiC bench path yields 0.45 nm K^{-1} . Common-mode expansion of the inter-craft baseline is not substituted for this internal optical-path coefficient.

C. Ten-Year GNC Propellant Consumption Budget

The primary mission interval is $T_{10} = 10.000$ yr. Both craft use EMI–BF₄ ionic-liquid electrospray propulsion with $I_{\text{sp}} = 3200.0$ s. The installed propellant capacities are 850.0 kg for the Lens Craft and 340.0 kg for the Sensor Craft. The propulsion ledger uses beginning-of-life maneuvering masses $m_{\text{L},0} = 4250.0$ kg and $m_{\text{S},0} = 1700.0$ kg for the two respective control plants.

TABLE XIII
SUN-EARTH L2 FORMATION-DISTURBANCE ENVIRONMENT

Disturbance	Magnitude	Frequency domain / characteristic	Principal GNC effect and accommodation
Differential SRP	$3.42 \times 10^{-8} \text{ m s}^{-2}$	Secular; $f < 10^{-4} \text{ Hz}$	Axial drift and transverse bias; feed-forward SRP model plus electrospray trim.
Sun-Earth tidal gravity	$\sim 4.7 \times 10^{-11} \text{ m s}^{-2}$ at 169.30 m	Halo-orbit harmonics; fundamental near $6.6 \times 10^{-7} \text{ Hz}$	Periodic baseline dilation and attitude coupling; ephemeris-resolved state propagation.
Residual outgassing	$S_a^{1/2} \leq 1.0 \times 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$	Broadband, 10^{-5} – 10^{-1} Hz ; 14.0-day decay	Stochastic translation and torque; augmented-state bias estimation and low-frequency thrust rejection.
C/SiC thermoelastic path	0.45 nm K^{-1}	Thermal transients, 10^{-4} – 10^{-2} Hz	Metrology phase bias; temperature feed-forward and in-flight zero-path calibration.
Mechanism and electrical jitter	Allocated from measured line spectra	0.1 Hz–100 kHz	Beam-pointing and phase modulation; electro-optic correction without spacecraft momentum exchange.

The annual differential-SRP increment follows directly from Eq. (97):

$$\begin{aligned}
 T_{\text{yr}} &= 365.25(86400 \text{ s}) = 31\,557\,600 \text{ s}, \\
 \Delta v_{\text{SRP, yr}} &= \Delta a_{\text{SRP}} T_{\text{yr}} \\
 &= (3.42 \times 10^{-8} \text{ m s}^{-2})(31\,557\,600 \text{ s}) \\
 &= 1.0792 \text{ m s}^{-1} \text{ yr}^{-1}, \\
 \Delta v_{\text{SRP, 10}} &= 10.792 \text{ m s}^{-1}.
 \end{aligned} \tag{101}$$

The outgassing compensation allocation is based on the conservative secular acceleration $a_{\text{outgas}} = 5.0 \times 10^{-9} \text{ m s}^{-2}$:

$$\begin{aligned}
 \Delta v_{\text{outgas, yr}} &= a_{\text{outgas}} T_{\text{yr}} = 0.1578 \text{ m s}^{-1} \text{ yr}^{-1}, \\
 \Delta v_{\text{outgas, 10}} &= 1.578 \text{ m s}^{-1}.
 \end{aligned} \tag{102}$$

Halo-orbit station keeping contributes $\Delta v_{\text{sk}} = 2.5000 \text{ m s}^{-1} \text{ yr}^{-1}$, or 25.000 m s^{-1} over ten years. Autonomous focal transitions carry separate ten-year allowances of 15.000 m s^{-1} for the Lens Craft and 25.000 m s^{-1} for the Sensor Craft.

The corresponding propellant expenditure is evaluated with the Tsiolkovsky relation

$$\Delta m = m_0 \left[1 - \exp\left(-\frac{\Delta v}{g_0 I_{\text{sp}}}\right) \right], \quad g_0 = 9.80665 \text{ m s}^{-2}. \tag{103}$$

For the cumulative allocations in Table XIV,

$$\begin{aligned}
 \Delta m_{\text{lens}} &= (4250.0 \text{ kg}) \left[1 - \exp\left(-\frac{52.370}{(3200.0)(9.80665)}\right) \right] \\
 &= 7.09 \text{ kg}, \\
 \Delta m_{\text{sensor}} &= (1700.0 \text{ kg}) \left[1 - \exp\left(-\frac{62.370}{(3200.0)(9.80665)}\right) \right] \\
 &= 3.38 \text{ kg}.
 \end{aligned} \tag{104}$$

The end-of-life propellant inventories and reserve fractions are

$$\begin{aligned}
 m_{\text{L, EOL}} &= 850.0 - 7.09 = 842.91 \text{ kg}, \quad \mathcal{R}_{\text{L, EOL}} = 99.17\%, \\
 m_{\text{S, EOL}} &= 340.0 - 3.38 = 336.62 \text{ kg}, \quad \mathcal{R}_{\text{S, EOL}} = 99.01\%.
 \end{aligned} \tag{105}$$

Both propulsion systems therefore retain more than 98% of installed propellant at the end of the ten-year primary mission.

Focal repositioning uses the normalized time $\tau = t/T_{\text{trans}}$ and the fifth-order minimum-jerk profile

$$\begin{aligned}
 s(\tau) &= 10\tau^3 - 15\tau^4 + 6\tau^5, \quad 0 \leq \tau \leq 1, \\
 z_{\text{ref}}(t) &= z_0 + \Delta z s(\tau).
 \end{aligned} \tag{106}$$

Its dimensionless velocity and acceleration functions are

$$\begin{aligned}
 s'(\tau) &= 30\tau^2 - 60\tau^3 + 30\tau^4, \\
 s''(\tau) &= 60\tau - 180\tau^2 + 120\tau^3, \\
 \max_{\tau \in [0,1]} s'(\tau) &= 1.875 \quad \text{at } \tau = 0.5, \\
 \max_{\tau \in [0,1]} |s''(\tau)| &= 5.7733 \quad \text{at } \tau = \frac{3 \pm \sqrt{3}}{6}.
 \end{aligned} \tag{107}$$

The boundary conditions $s(0) = 0$, $s(1) = 1$, $s'(0) = s'(1) = 0$, and $s''(0) = s''(1) = 0$ eliminate endpoint velocity and acceleration steps. Physical limits follow as $|\dot{z}|_{\text{max}} = 1.875|\Delta z|/T_{\text{trans}}$ and $|\ddot{z}|_{\text{max}} = 5.7733|\Delta z|/T_{\text{trans}}^2$.

D. Two-Tier Hybrid Control Architecture

The relative-state estimator fuses heterodyne range, DWS pitch and yaw, optical-clock transfer, and inertial propagation. Complementary filters divide the range and pointing residuals at the 0.1-Hz crossover. Low-frequency errors are assigned to multimodal EMI-BF₄ ionic-liquid electrospray microthrusters with a commanded range of 0.1–150.0 μN and $I_{\text{sp}} = 3200.0 \text{ s}$. The thrust allocator removes secular SRP, gravity-gradient, and outgassing biases while maintaining the electro-optic stage near midrange.

For $0.1 \text{ Hz} \leq f \leq 100 \text{ kHz}$, an integrated 8×8 InP/InGaAs phase-shifter array corrects optical-path and beam-pointing jitter without imparting mechanical momentum. Each channel obeys

$$\begin{aligned}
 \Delta \phi_{\text{EO}}(V) &= \frac{\pi}{V_{\pi}} [V_{\text{bias}}(t) - V_0], \\
 V_{\pi} &= 3.6 \text{ V}, \quad V_0 = 3.8 \text{ V},
 \end{aligned} \tag{108}$$

with $3.8 \leq V_{\text{bias}} \leq 7.4 \text{ V}$, response time no greater than 5 ns, and device modulation bandwidth above 40 GHz. The control band is limited to 100 kHz by metrology latency and noise shaping rather than device speed.

TABLE XIV
TEN-YEAR CUMULATIVE GNC DELTA-V BUDGET

Contributor	Lens annual (m/s/yr)	Lens 10-yr (m/s)	Sensor annual (m/s/yr)	Sensor 10-yr (m/s)
Halo-orbit station keeping	2.5000	25.000	2.5000	25.000
Differential solar-radiation-pressure rejection	1.0792	10.792	1.0792	10.792
Spacecraft outgassing compensation	0.1578	1.578	0.1578	1.578
Autonomous focal repositioning allowance	—	15.000	—	25.000
Total cumulative allocation	3.7370 recurring	52.370	3.7370 recurring	62.370

For measured range error $e_r = r_{\text{cmd}} - r_{\text{meas}}$, the complementary command split is

$$\begin{aligned} u_{\text{thr}}(s) &= G_{\text{thr}}(s)H_{\text{LP}}(s)e_r(s), \\ u_{\text{EO}}(s) &= G_{\text{EO}}(s)H_{\text{HP}}(s)e_r(s), \end{aligned} \quad (109)$$

where $H_{\text{LP}} + H_{\text{HP}} = 1$ and both paths include saturation and anti-windup logic. The electro-optic array provides a calibrated $\pm 10 \mu\text{m}$ equivalent optical-path stroke. Persistent stroke utilization above 80% transfers bias to the thruster loop; QPD loss of lock freezes the phase command and initiates a low-rate spiral reacquisition rather than integrating an unobservable error.

E. Fault-Tolerant $N - k$ Power Derating and Focal Repositioning

Let k be the number of unavailable LANR modules. The nominal plant output, entropy-debt demand, and operating margin are

$$\begin{aligned} P_{\text{cluster},0} &= 906.0726 \text{ kW}, \\ P_{\text{debt},0} &= 906.0000 \text{ kW}, \\ \Delta P_0 &= P_{\text{cluster},0} - P_{\text{debt},0} = +72.60 \text{ W}. \end{aligned} \quad (110)$$

The $N - k$ outage dynamics are

$$\begin{aligned} P_{\text{net}}(k) &= (1786 - k)(507.3195 \text{ W}), \\ \Delta P_{\text{net}}(k) &= P_{\text{net}}(k) - P_{\text{debt},0}, \\ P_{\text{seed}}(k) &= \min[P_{\text{net}}(k) - P_{\text{reserve}}, P_{\text{debt},0}], \end{aligned} \quad (111)$$

where $P_{\text{reserve}} \geq 0$ is the transient and bus-stability reserve. For $k = 1$,

$$P_{\text{net}}(1) = 905.5653 \text{ kW}, \quad \Delta P_{\text{net}}(1) = -0.4347 \text{ kW}. \quad (112)$$

The fast command-plane response converts the positive loss magnitude $\Delta P_{\text{loss}} = \max(-\Delta P_{\text{net}}, 0)$ into an integer bit-overflow decrement,

$$\begin{aligned} \Delta N_{\text{hw}} &= \left\lfloor \frac{\Delta P_{\text{loss}}}{P_{\text{bit}}} \right\rfloor, \\ P_{\text{bit}} &= 8.9506 \times 10^{-4} \text{ W bit}^{-1}. \end{aligned} \quad (113)$$

Thus, a single-module outage commands $\Delta N_{\text{hw}} = 485\,665$ hardware decrement counts before the continuously valued seed command is quantized. Here P_{bit} is the power-equivalent scale of the flight controller's finite register command, not the thermodynamic cost of one microscopic boundary state. The

physical boundary-state excess remains governed by Eqs. (1) and (69), for which the effective coefficient is $1.20117 \times 10^{-39} \text{ W bit}^{-1}$.

The normalized power fraction and continuous seed-mass command are

$$\begin{aligned} \rho_P(k) &= \frac{P_{\text{seed}}(k)}{P_{\text{debt},0}}, \\ M_{\text{seed}}[P_{\text{seed}}(k)] &= M_{\text{seed},0} \rho_P(k), \\ \Delta N_{\text{cmd}}(k) &= \lfloor \Delta N_0 \rho_P(k) \rfloor. \end{aligned} \quad (114)$$

Equivalently, when $P_{\text{net},0}$ denotes the 906.0000-kW power allocated to the nominal seed rather than the 906.0726-kW plant output,

$$M_{\text{seed}}(P_{\text{net}}) = M_{\text{seed},0} \frac{P_{\text{net}}}{P_{\text{net},0}}. \quad (115)$$

For zero additional reserve, a single-module outage commands $M_{\text{seed}} = 9.99520 \times 10^{-7} M_{\odot}$ and $f = 169.381 \text{ m}$. At the science-safe minimum $M_{\text{seed},\text{min}} = 10^{-7} M_{\odot}$, the seed demand is 90.60 kW. The $k = 1607$ state leaves 179 active modules producing 90.8082 kW; the controller assigns 90.60 kW to the seed and retains 208.2 W as reserve. Focal distance follows

$$f(k) = f_0 \frac{M_{\text{seed},0}}{M_{\text{seed}}(k)} = f_0 \frac{P_{\text{debt},0}}{P_{\text{seed}}(k)}, \quad (116)$$

and increases continuously from $f_0 = 169.30 \text{ m}$ at $10^{-6} M_{\odot}$ to $f_{\text{max}} = 1\,692.99 \text{ m}$ at $10^{-7} M_{\odot}$.

The Lens Craft broadcasts an authenticated tuple containing the module-validity mask, P_{seed} , ΔN_{cmd} , seed-mass command, and command epoch. After independent consistency checks, the Sensor/Focal Craft sets

$$\mathbf{r}_{\text{target}}(t) = \mathbf{r}_{\text{Lens}}(t) + f[M_{\text{seed}}(t)]\hat{\mathbf{z}}_{\text{optical}}(t). \quad (117)$$

For a commanded axial change Δz , the Sensor/Focal Craft executes the fifth-order minimum-jerk reference of Eq. (106). Transition duration is selected from the velocity and acceleration limits in Eq. (107) so that commanded thrust, propellant, line-of-sight rate, and electro-optic stroke remain within limits. Heterodyne range and DWS remain closed throughout the maneuver. Invalid or stale power telemetry, disagreement between the reported power and register count, loss of metrology, or $P_{\text{seed}} < 90.60 \text{ kW}$ commands an orderly field quench rather than extrapolation below the qualified seed envelope.

F. Non-Paraxial Wide-Angle Wavefront Optics & DAC Pre-Distortion

Tracking commands with $5^\circ < \Theta_{\text{tilt}} \leq 15^\circ$ use an exact eikonal path model rather than the paraxial phase ramp of Eq. (34). For aperture coordinates (x, y) , commanded angular vector $\Theta = (\Theta_x, \Theta_y)$, and nominal focal distance f_0 , define

$$\begin{aligned} a(x, y; \Theta) &= x \sin \Theta_x + y \sin \Theta_y, \\ r^2 &= x^2 + y^2, \\ W(x, y; \Theta) &= \sqrt{f_0^2 + r^2 - 2f_0 a(x, y; \Theta)} - f_0. \end{aligned} \quad (118)$$

Expansion of Eq. (118) in powers of r/f_0 gives

$$W = -a + \frac{r^2 - a^2}{2f_0} + \frac{ar^2 - a^3}{2f_0^2} + \mathcal{O}\left(\frac{r^4}{f_0^3}\right). \quad (119)$$

In the commanded steering plane, $a = r \sin \Theta_{\text{tilt}}$, so that $r^2 - a^2 = r^2 \cos^2 \Theta_{\text{tilt}}$. Applying the compensating phase $\phi_{\text{eikonal}} = -k_0 W$ produces the high-order command

$$\begin{aligned} \phi_{\text{eikonal}}(x, y) &= k_0 \left[a - \frac{r^2}{2f_0} \cos^2 \Theta_{\text{tilt}} \right. \\ &\quad \left. - \frac{r^2 a}{2f_0^2} \cos^2 \Theta_{\text{tilt}} + \mathcal{O}\left(\frac{r^4}{f_0^3}\right) \right], \\ k_0 &= \frac{2\pi}{\lambda_0}, \quad \Theta_{\text{tilt}} = \sqrt{\Theta_x^2 + \Theta_y^2}. \end{aligned} \quad (120)$$

The cubic contribution $r^2 a \cos^2 \Theta_{\text{tilt}} / (2f_0^2)$ supplies the pre-distortion that cancels the dominant off-axis coma. The phase map is evaluated at every emitter coordinate (x_m, y_n) and combined with the tensor-correction phase before voltage quantization.

Near collector saturation, the voltage-dependent collector junction capacitance is

$$C_c(V_D) = \frac{C_{c0}}{(1 + V_D/V_{\text{bi}})^{m_c}}, \quad (121)$$

where C_{c0} is the zero-bias capacitance, V_{bi} is the built-in potential, and m_c is the junction grading coefficient. The inverse calibration that removes the resulting nonlinear phase delay is represented by the fourth-order transfer law

$$\begin{aligned} V_{\text{DAC}}(\theta_{m,n}) &= V_0 + c_1 \theta_{m,n} + c_2 \theta_{m,n}^2 \\ &\quad + c_3 \theta_{m,n}^3 + c_4 \theta_{m,n}^4, \\ V_0 &= 3.800 \text{ V}, \quad 3.800 \leq V_{\text{DAC}} \leq 7.400 \text{ V}. \end{aligned} \quad (122)$$

The coefficients c_1 through c_4 are obtained from cryogenic per-channel phase-versus-voltage calibration and stored with the active wavefront solution. The resulting voltage is converted to the 16-bit register code through Eq. (37).

Across the complete steering envelope, the pre-distorted command maintains $\sigma_\phi \leq 9.8 \times 10^{-5}$ rad and therefore $S \geq 0.99999990396 > 0.999999980$. The wide-angle controller reverts to the calibrated paraxial map below 5° and rejects commands above 15° .

G. Diffraction-Limited Optical Baseline

At the reference impact radius $r_0 = 1.000$ m, the unobscured synthetic lens has an effective aperture

$$D_{\text{eff}} = 2r_0 = 2.000 \text{ m} \quad (123)$$

at the baseline focal distance $f_0 = 169.30$ m. For monochromatic observations at $\lambda = 500$ nm, the Rayleigh angular resolution is

$$\begin{aligned} \theta_{\text{res}} &= 1.22 \frac{\lambda}{D_{\text{eff}}} \\ &= 1.22 \frac{500 \times 10^{-9} \text{ m}}{2.000 \text{ m}} \\ &= 3.05 \times 10^{-7} \text{ rad} \\ &= 0.0629 \text{ arcsec}. \end{aligned} \quad (124)$$

This value defines the ideal reconstruction-grid scale before detector sampling, formation error, finite bandwidth, and residual metric-phase error are folded into the system point-spread function (PSF).

The on-caustic coherent amplitude magnification is specified by the circumference-to-wavelength ratio

$$\begin{aligned} \mu_{\text{amp}} &= \frac{2\pi r_0}{\lambda} \\ &= \frac{2\pi(1.000 \text{ m})}{500 \times 10^{-9} \text{ m}} \\ &= 1.2566 \times 10^7. \end{aligned} \quad (125)$$

Expressed as an astronomical magnitude gain,

$$\Delta m = 2.5 \log_{10} \mu_{\text{amp}} = +17.75 \text{ mag}. \quad (126)$$

The quoted gain is the ideal peak caustic value; science-product calibration shall apply the finite-source, spectral-bandpass, detector, and caustic-centering transfer functions.

H. Wavefront Quality and Strehl Acceptance

For zero-mean rms phase error σ_ϕ , the Maréchal approximation gives

$$S = \exp(-\sigma_\phi^2). \quad (127)$$

Using the closed-loop phase-jitter bound $|\delta\phi| \leq 5.05 \times 10^{-5}$ rad as the conservative rms allocation,

$$\begin{aligned} S_{\text{min}} &= \exp\left[-(5.05 \times 10^{-5})^2\right] \\ &= 0.99999999745 \\ &> 0.99999999. \end{aligned} \quad (128)$$

Thus, phase jitter alone contributes less than 2.56×10^{-9} fractional peak-intensity loss. The reconstructed PSF shall nevertheless include measured amplitude imbalance, detector pixel response, spectral dispersion, and formation-state covariance rather than treating Eq. (128) as the complete optical error budget.

TABLE XV
NON-PARAXIAL STEERING PARAMETERS

Steering state Θ_{tilt}	Target phase shift $\theta_{m,n}$ (rad)	DAC code (16 bit)	Drive voltage V_{DAC} (V)	Residual phase error σ_ϕ (rad)	Maréchal Strehl ratio $S = \exp(-\sigma_\phi^2)$
0.0°, boresight	0.0000	0	3.8000	1.2×10^{-6}	0.999999999999
2.5°, paraxial	1.5708 ($\pi/2$)	14563	4.6000	4.5×10^{-6}	0.999999999980
5.0°, boundary	3.1416 (π)	29127	5.4000	1.2×10^{-5}	0.999999999856
7.5°, non-paraxial	4.7124 ($3\pi/2$)	43690	6.2000	2.8×10^{-5}	0.999999999216
10.0°, wide angle	6.2832 (2π)	52428	6.6800	4.9×10^{-5}	0.999999997599
12.5°, extended	7.8540 ($5\pi/2$)	59710	7.0800	7.3×10^{-5}	0.999999994671
15.0°, maximum	9.4248 (3π)	65535	7.4000	9.8×10^{-5}	0.999999990396

I. Regularized Modified Richardson–Lucy Reconstruction

Let $\mathbf{y} \geq 0$ be the background-subtracted focal-plane photon counts, $\mathbf{x} \geq 0$ the unknown celestial intensity distribution, $\mathbf{H}$ the spatially variant caustic-response operator, and $\mathbf{b} \geq 0$ the estimated dark-current and stray-light background. The forward Poisson model is

$$\mathbf{y} \sim \text{Poisson}(\mathbf{H}\mathbf{x} + \mathbf{b}). \quad (129)$$

The response operator shall be generated from the measured wavelength-dependent PSF and normalized such that $\mathbf{H}^\top \mathbf{1}$ captures edge truncation and nonuniform exposure. Starting from a strictly positive flat-field or back-projected estimate $\mathbf{x}^{(0)}$, iteration k forms

$$\begin{aligned} \mathbf{r}^{(k)} &= \frac{\mathbf{y}}{\mathbf{H}\mathbf{x}^{(k)} + \mathbf{b} + \varepsilon}, \\ \mathbf{c}^{(k)} &= \frac{\mathbf{H}^\top \mathbf{r}^{(k)}}{\mathbf{H}^\top \mathbf{1} + \varepsilon}, \\ \tilde{\mathbf{x}}^{(k+1)} &= \mathbf{x}^{(k)} \odot \mathbf{c}^{(k)}, \end{aligned} \quad (130)$$

where division and $\odot$ are elementwise and $\varepsilon > 0$ prevents division by zero. A nonnegative Laplacian-regularized, relaxed update suppresses caustic ringing and noise amplification:

$$\mathbf{x}^{(k+1)} = \mathcal{P}_+[(1 - \beta)\mathbf{x}^{(k)} + \beta\tilde{\mathbf{x}}^{(k+1)} - \tau \mathbf{L}^\top \mathbf{L}\mathbf{x}^{(k)}], \quad (131)$$

where $0 < \beta \leq 1$ is the relaxation factor, $\tau \geq 0$ is selected from calibration data, $\mathbf{L}$ is the discrete gradient operator, and $\mathcal{P}_+$ projects negative samples to zero. After projection, each iterate is flux-normalized to the background-corrected count estimate:

$$\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k+1)} \frac{\max(\mathbf{1}^\top (\mathbf{y} - \mathbf{b}), 0)}{\mathbf{1}^\top \mathbf{H}\mathbf{x}^{(k+1)} + \varepsilon}. \quad (132)$$

Iteration terminates when both the relative image change and relative Poisson deviance change fall below their calibrated tolerances, or when the fixed iteration ceiling is reached:

$$\frac{\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2}{\|\mathbf{x}^{(k)}\|_2 + \varepsilon} < \epsilon_x, \quad \frac{|\mathcal{D}_{k+1} - \mathcal{D}_k|}{\mathcal{D}_k + \varepsilon} < \epsilon_D. \quad (133)$$

Reconstruction products shall carry the PSF version, regularization parameters, iteration count, residual deviance, photometric normalization, and propagated formation/metrology covariance needed for reproducibility.

VII. CONCLUSION & SYSTEMS VERIFICATION GATE

The SGLT preliminary-design baseline is accepted only when the analytical, numerical, hardware, and integrated-test products listed in Table XVI are configuration controlled and independently reproduced. The matrix cross-links the four governing test benches: `shbt-exotic` for metric and lens physics, `shbt-qc` for quantum-state and cryogenic transducer control, `shbt-cf` for LANR power and heat rejection, and `shbt-precision` for deterministic computing, spacecraft resources, formation metrology, and optical reconstruction. A bidirectional arrow denotes a required exchange of test vectors and acceptance evidence between benches.

Closure of the matrix establishes traceability from the holographic seed law through power generation, heat rejection, metric projection, deterministic fault containment, formation control, focal-plane acquisition, and reconstructed science data. With every verification gate passed and its evidence placed under configuration control, the SGLT architecture constitutes a closed integrated engineering blueprint suitable for progression from preliminary design into detailed subsystem qualification and end-to-end verification.

TABLE XVI
SGLT MASTER SYSTEM VERIFICATION MATRIX

Subsystem	Governing parameter	Theoretical target	Verified baseline value	Primary / cross-linked test bench
Metric seed synthesis	Seed mass and state excess, Eq. (1)	$M_{\text{seed}} = 10^{-6} M_{\odot}$	$1.0000(10) \times 10^{-6} M_{\odot}$	shbt-exotic ↔ shbt-qc
Gravitational lens	Focal distance, Eq. (3)	$f = 169.3 \text{ m at } r_0 = 1 \text{ m}$	169.30(19) m	shbt-exotic ↔ shbt-precision
Boundary stability	Eigenvector detuning, Eq. (4)	$ \delta\mu \leq 10^{-12}$	1.0×10^{-12} limit	shbt-qc ↔ shbt-exotic
Transducer phasing	Phase jitter, Eq. (128)	$ \delta\phi \leq 5.05 \times 10^{-5} \text{ rad}$	$S_{\text{min}} = 0.99999999745$	shbt-qc ↔ shbt-precision
Acoustic matching	Reflected fraction, Eq. (52)	$R < 0.2\%$	$R = 0.1888\%$	shbt-qc ↔ shbt-cf
LANR generation	Continuous net power, Eq. (73)	$P_{\text{net}} \geq 906.00 \text{ kW}$	906.0726 kW, 1786 modules	shbt-cf ↔ shbt-qc
Heat rejection	Radiator area at 600 K, Eq. (75)	Reject 4.6188 MW	683.17 m ² effective area	shbt-cf ↔ shbt-qc
Cryogenic boundary	Areal thermal resistance, Eq. (78)	$R''_{\text{th}} < 7.04 \times 10^{-6}$	Requirement closed by active control	shbt-qc ↔ shbt-cf
Spacecraft resources	Mass, power, propulsion, and telecom, Table II	Dry/wet mass and bus-load closure	5 370/6 560 kg; 910.8726 kW	shbt-precision ↔ shbt-cf
SRAM and ECC	Frame size and SECDED decision, Eqs. (81) and (84)	2112 byte; correct 1 bit, detect 2 bit	640 + 1472 = 2112 byte; (72, 64) SECDED	shbt-precision ↔ shbt-qc
Safety interlock	SIMD decision and shunt, Eq. (87)	$t_{\text{shunt}} < 2.50 \text{ ns}$	1.50 ns decision; 1.00 ns margin	shbt-precision ↔ shbt-qc
Recovery control	Return to phase-valid state, Eq. (89)	$t_{\text{recover}} \leq 120.0 \text{ ns}$	110 ns hold plus 10 ns recovery	shbt-precision ↔ shbt-qc
Formation flight	Relative-position error, Eq. (6)	$\ \delta\mathbf{r}\ _{3\sigma} \leq 1.00 \text{ nm}$	1.00 nm allocation	shbt-precision ↔ shbt-exotic
Laser metrology	Phase and range noise, Eqs. (93) and (94)	$S_{\phi}^{1/2} \leq 2.0 \mu\text{rad}/\sqrt{\text{Hz}}$	$S_r^{1/2} \leq 0.1694 \text{ pm}/\sqrt{\text{Hz}}$	shbt-precision ↔ shbt-exotic
DWS attitude sensing	QPD angular knowledge, Eq. (95)	$\sigma_{\theta} \leq 0.015 \mu\text{rad}$	15 nrad per-axis allocation	shbt-precision ↔ shbt-exotic
Hybrid GNC	Thruster/phase-shifter split, Eq. (109)	0.1-Hz crossover	0.1–150 μN and 0.1 Hz–100 kHz	shbt-precision ↔ shbt-qc
Power-fault response	$N - k$ seed derating, Eqs. (111)–(116)	Stable through $k = 1607$	$10^{-7} M_{\odot}$ at 1.692999 km	shbt-cf ↔ shbt-precision
Boundary-scale map	Congestion/exclusion factor, Eq. (19)	Separate boundary and bulk domains	$\chi_{\text{SHBT}} = 1.002341 \times 10^3$	shbt-exotic ↔ shbt-qc
Optical resolution	Rayleigh angle, Eq. (124)	$\lambda = 500 \text{ nm}$, $D_{\text{eff}} = 2 \text{ m}$	$3.05 \times 10^{-7} \text{ rad}$ (0.0629 arcsec)	shbt-precision ↔ shbt-exotic
Image reconstruction	Regularized modified RL, Eqs. (130)–(133)	Nonnegative, flux-normalized convergence	Dual-tolerance stopping and fixed ceiling	shbt-precision ↔ shbt-exotic
Integrated system gate	Configuration-controlled evidence package	All subsystem gates passed	Four-bench evidence concurrence	shbt-exotic ↔ shbt-qc ↔ shbt-cf ↔ shbt-precision